# A Sufficient Statistic for Algorithmic Feedback: What Prices Reveal About Trader Groups, and Who Needs the Rest

Pratik Niroula* Prashant Niroula † Sujan Khatiwada‡

Working paper, core version: 12 November 2026

**Abstract**

In a market with algorithmic feedback traders, the feedback kernel, the response of total feedback flow to past prices, is a sufficient statistic: for any insider and causal dealer rule, populations with the same kernel generate identical prices, profits and equilibria, and prices identify nothing finer. Learning group composition costs data of order $(\Delta\eta)^{-(4K-2)}$, in the closed loop, in equilibrium and near instability. With position limits, calm prices reveal only the kernel, and only stress reveals the rest. In a pre-registered out-of-sample test on Taiwan's tagged exchange flows (2019–2023), prices improve forecasts of foreign investors' flow as the registered criterion requires, but a pre-specified check attributes the gain to prices revealing past flow rather than to feedback. A registered test of the position-limit prediction on Binance perpetual futures is inconclusive. Core statements are machine-checked in Lean 4.

**AI-use disclosure.** Claude, ChatGPT/Codex and Gemini provided material assistance with research development, code, numerical checks, manuscript drafting and internal critique, including the restructuring of this core version. Claude and ChatGPT/Codex developed and checked Lean proofs. Human authors remain responsible for verifying the content and its scholarly contribution before submission. AI checks do not constitute independent review. The authors have verified all mathematical proofs, Lean 4 formalizations and empirical calculations, and take full responsibility for the content of the manuscript.

*Department of Computer Information Science, Minnesota State University, Mankato, MN 56001, USA. pratik.niroula@mnsu.edu.

†Faculty of Management, Tribhuvan University, Kirtipur, Nepal. prashantniroula20@gmail.com.

‡Department of Electrical, Computer and Robotics Engineering, Minnesota State University, Mankato, MN 56001, USA. sujan.khatiwada@mnsu.edu.

**Relation to the full working paper.** This is a shortened version, organized around one result. Every result stated here also appears in the full working paper ("What Prices Cannot Tell a Market Supervisor", 5 November 2026), with the same proof. The exceptions are Proposition 3.2, Proposition 6.6, Supplement S5.1 and all of Section 7, which are new to this version and have not appeared elsewhere. The full working paper calls the trader groups "cohorts". Results not needed for the main argument are omitted, among them account-level capital, general memory, spectral blindness and the empirical tests, which this version only summarizes. Section 8 says where they are.

**Keywords:** algorithmic trading, feedback trading, sufficient statistics, aggregation, identification, market microstructure, supervision, disclosure.

**JEL:** C32, D83, G14, G18.

**Code and proofs.** All code, Lean proofs and result files are public at https://github.com/Pratikn03/Fianince.

# 1 Introduction

Much of today's trading is done by programs that read recent prices, decide what to trade, and by trading move the prices they read. Trend-following funds, execution algorithms, market-making programs and, more recently, AI trading agents all work this way. When such a market misbehaves, supervisors ask who was trading and how fast those traders react. This paper asks a prior question: what does a market's price record reveal about its population of feedback traders, and for which decisions is that enough?

**The answer in one sentence.** Prices reveal the market's *feedback kernel*: the response, $m$ periods later, of total feedback flow to a unit move in the price signal. That kernel is all that prices identify, and it is all that anyone trading on public information needs. Group composition, meaning who the groups are and how strongly each trades, matters only for actions aimed at a particular group, and learning it from prices has a known and steep price.

**The model.** The market has a price and $K$ groups of traders. Each group keeps an exponentially weighted average of past prices, with memory rate $r_i$, and trades in proportion to it with weight $w_i$. A desk that tracks a trend with a Kalman filter, or that learns by regularized online gradient descent, behaves exactly this way (Lemma 2.1). Total feedback flow is then a convolution of past prices with the kernel

$$k_m = \sum_i w_i r_i^m, \qquad m = 0, 1, 2, \ldots,$$

the moments of the *merged measure* $M$, which puts weight $w_i$ at rate $r_i$. The paper rests on three results about this one object.

(1) *The kernel is sufficient* (Theorem 3.1 and Proposition 3.2). Add a value, noise traders, an insider and a dealer, each following an arbitrary causal rule, which covers Kyle, inventory-averse, learning and misspecified dealers. Two trader populations with the same kernel then produce the same price and order-flow paths, the same insider trades and profits and the same dealer inventories, realization by realization. The two games have the same strategy sets and payoffs, and hence the same equilibria. The proof needs only that aggregate feedback flow is a fixed linear function of past public signals, so the result holds for any linear trader population, not only exponential memories.

(2) *The kernel is all that prices identify* (Theorem 3.3). Two populations give identical price responses to every trade sequence if and only if their merged measures coincide.

Groups with the same memory cannot be separated however long the record. Groups with different memories can be separated in principle.

(3) *Learning more has a known price* (Theorems 5.1, 5.2 and 5.4). To separate $K$ groups whose memories lie within $\Delta\eta$ of each other takes of order $(\Delta\eta)^{-(4K-2)}$ experiments, and Prony's method attains this order. Waiting instead of trading does not help: passive observation needs at least the same order of observations.

**The results survive equilibrium.** In the base model the price impact and the persistence of mispricing are primitives. Section 4 derives both from competitive dealers who dislike holding inventory. The equilibrium price is then the same closed loop, path by path, with effective impact and persistence and with the memory rates unchanged (Theorem 4.2). So the kernel remains the sufficient statistic, and the cost of going beyond it keeps its order. The supplement adds a strategic insider with short-lived information. It also shows that, with positive feedback, liquidity disappears beyond a threshold in dealer risk aversion that falls with the feedback intensity.

**What the kernel misses: limits.** Real feedback traders have position limits, margin and risk budgets. Section 7 gives a two-layer theory for traders whose positions saturate.

- The sufficient object becomes a capacity-weighted measure over memory and leverage, of which the kernel is one linear functional (Theorem 7.1).
- Calm-market prices reveal the kernel and nothing more. Under hard limits, two populations with the same kernel give identical prices until some trader reaches its limit (Theorem 7.2). Calm data therefore carry no information about how close traders are to their limits, and Gaussian data carry exponentially little. Under smooth limits, the information grows like the sixth power of the probe (Proposition 7.3).
- Stress reveals everything that matters. Responses to probes of all sizes identify the net capacity at every memory and leverage level, which is exactly the equivalence class (Theorem 7.4).
- The results extend to traders who deleverage or liquidate outside their bands (Proposition 7.5). An informed dealer who prices next-period flow leaks leverage information into calm prices, but only at the scale of the probability of a breach (Proposition 7.6).

**Supervision splits in two.** A market-wide action, such as a uniform cut in activity to meet a ceiling on the feedback budget, depends on the population only through the kernel. A targeted action, such as halting the largest group, does not. Two populations whose

price responses differ only by order $(\Delta\eta)^3$ can need different halts (Proposition 6.1). For such decisions, flow reported separately by trader class turns each group into an ordinary regression, with no penalty as the groups' memories approach each other (Proposition 6.2). It cuts the number of experiments needed by a factor of order $(\Delta\eta)^{-(4K-2)}$ (Theorem 6.3).

## 1.1 What is known and what is new

The central idea of this paper, that an aggregate is determined by a distribution of individual coefficients and not by the individuals, is old. We state its precedents first, because they fix what we can claim.

**Aggregation of dynamic models.** Granger (1980) showed that aggregating many first-order autoregressions with different coefficients produces a process whose dynamics are governed by the distribution of the coefficients; with enough mass near one the aggregate has long memory. Zaffaroni (2004) develops the theory for large economies. Recovering the coefficient distribution from aggregate data is known as the disaggregation problem (Celov et al., 2010). Our merged measure $M$ plays the role of that coefficient distribution, and Theorem 3.3(i) is the corresponding identification statement for a closed feedback loop. Forni and Lippi (1997) develop the general theory of what aggregate dynamics reveal about heterogeneous individual dynamics, and Pesaran and Chudik (2014) the large-panel version.

**Sufficient statistics for aggregate dynamics.** In macroeconomics, Auclert et al. (2024) show that the matrix of intertemporal marginal propensities to consume is a sufficient statistic for the output response to fiscal policy, whatever the household heterogeneity behind it. The kernel $k$ plays the same role for a market with feedback traders. It is the response of aggregate demand to past prices, and it summarizes every heterogeneity that matters for prices.

**Observational equivalence.** Sargent (1976) showed that two macroeconomic theories with opposite policy implications generate the same reduced form, so that data from one policy regime cannot decide between them. Theorem 3.1 is a result of this kind for trader populations, and Section 6 draws the policy consequence: which supervisory actions the equivalence class supports, and which need data from outside it.

**Mixture estimation.** The order $(\Delta\eta)^{-(4K-2)}$ is the known minimax rate for finite mixtures with $K$ near-colliding components (Heinrich and Kahn, 2018; Wu and Yang, 2020),

and super-resolution theory gives the same conditioning for clustered exponentials (Moitra, 2015; Batenkov et al., 2021; Katz et al., 2023). We do not claim the exponent.

**Heterogeneous horizons under impact.** Leclère and Rosenbaum (2026) study a linear mean-field model in which agents with heterogeneous forecast horizons trade under market impact, and in which impact depends on aggregate positions rather than on trader identities. Their object is equilibrium and price regularity, not what an observer can identify or what supervision needs.

**Feedback trading and insiders.** Positive-feedback traders and their effect on prices go back to De Long et al. (1990). Barberis et al. (2015, 2018) model extrapolators with exponentially weighted memories, and Brock and Hommes (1998) model heterogeneous rules. The insiders and dealers covered by Section 3 include those of Kyle (1985) and of inventory models in the tradition of Ho and Stoll (1981) and Grossman and Miller (1988).

**What is new.** Five things, each of which uses the market setting.

(a) *Sufficiency inside a strategic market.* The aggregation literature concerns the statistical law of an aggregate series. Theorem 3.1 is a statement about games: it holds path by path for every insider strategy and every causal dealer rule, so it transfers to every equilibrium concept defined from payoffs and public histories. Part (iv) shows where it stops, since group-level quantities are not invariant.

(b) *Memories are zeros of the closed loop.* In mixture and open-loop problems the unknown rates are locations or poles of the observed law. In a feedback market they are transmission zeros of the price response (Theorem 5.3). That decides which estimators are valid: fitting poles to prices finds the closed-loop eigenvalues and misses the memories. Because the closed loop is minimum phase, passive data face at least the same barrier as active probing.

(c) *The results hold in equilibrium.* With inventory-averse dealers who price the feedback traders' predictable flow, the equilibrium is still the closed loop, with unchanged memories (Theorem 4.2).

(d) *Limits are invisible until reached, and stress reveals them exactly.* With saturating traders, the sufficient object is a measure over memory and leverage. Calm-market prices reveal only its linear part, and stress identifies all of it (Section 7). The last step of the stress result rests on a known uniqueness theorem for networks with one hidden layer (Sussmann, 1992); the reduction from prices to that setting is ours.

(e) *A supervisory split with a price.* The kernel suffices for market-wide actions and not for targeted ones, and class-tagged disclosure removes the cost of the targeted information (Propositions 6.1 and 6.2, Theorem 6.3). A group that can mimic another group's memory also caps what any price-based targeting can achieve, whatever the amount of data (Proposition 6.6).

The induction behind Theorem 3.1(iii) is short. Its content is not the induction but the object it isolates. The kernel is identified from prices, it is sufficient for every price-based outcome, and it is insufficient exactly for the targeted decisions supervisors care about.

**Machine checking.** The core statements are checked in Lean 4 against Mathlib with standard axioms only. The checks cover the sufficiency theorem (pathwise, for arbitrary causal rules), the $K$-group lower and upper bounds, the minimum-phase identification, the finite-sample passive bound, the reduced form of the inventory equilibrium, and the reduced form and best response in the insider equilibrium. Each statement says what is checked and what is proved only on paper; Supplement **??** lists the declarations.

**Outline.** Section 2 sets out the model. Section 3 proves that the kernel is sufficient and that it is what prices identify. Section 4 shows that both hold in equilibrium. Section 5 prices the information beyond the kernel. Section 6 shows which supervisory decisions need it and what disclosure buys. Section 7 turns to saturating traders. Section 8 states the limits, and Section 9 concludes.

# 2 Model and motivating examples

**The model in words.** The market has one price and $K$ groups of traders. Each period three things happen. First, the price moves: it drifts back toward its fundamental value, and it is pushed by the groups' trades, by any test trade and by news. Second, each group updates its view by mixing the newest price into its running average, with weight $\eta_i$ on the newest price and $1 - \eta_i$ on its old view. Third, each group trades in proportion to its view, with intensity $q_i$. A group with a large $\eta_i$ has a short memory and reacts quickly, like a short-horizon trend follower; a group with a small $\eta_i$ has a long memory and reacts slowly. The equations below write this down.

Time is discrete. Let $p_t$ be the price, $p^*$ the fundamental value and $x_t = p_t - p^*$ the pricing error. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, an imposed primitive rather than an equilibrium quote, and arbitrage pulls the error toward

zero with persistence $a \in [0, 1)$:

$$x_t = ax_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + \varepsilon_t, \tag{1}$$

where $v_t$ is an executed probe quantity and $\varepsilon_t$ a price innovation (news). Group $i$ holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1 - \eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0, 1). \tag{2}$$

The ordering is *price first*: the price responds to the probe, then groups update. Write $r_i = 1 - \eta_i$ for the memory rate and $w_i = q_i \eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j \geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z - a)D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \sum_j h_j = \kappa/(1 - S)$, where $S = a + \kappa \sum_i q_i$ is the feedback budget. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = Jz_{t-1} + bv_t$, where, because groups update on the same period's price, the probe enters every state on impact through

$$b = \kappa\, (1, \eta_1, \ldots, \eta_K)^\top, \tag{5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{6}$$

In the *positive economy* ($q_i \geq 0$), $J$ is entrywise nonnegative, its price row sums to $S$, and the economy is Schur stable iff $S < 1$.

**Lemma 2.1** (Two motivating examples: filtering and learning)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as*

$\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ *with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0,1). \tag{7}$$

*(ii)* Gradient memory. *Online gradient descent with $L_2$ regularization $\gamma \in (0,1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and $q = \alpha H/\eta$. A tabular value update with learning rate $\eta$ at a fixed state has the same form; general reinforcement-learning policies need not.*

**Why exponential memory, and what is assumed.** Exponential memory is the oldest model of expectations in economics: adaptive expectations (Cagan, 1956; Nerlove, 1958) and geometric distributed lags (Koyck, 1954), and Muth (1960) showed that exponentially weighted forecasts are optimal for a random walk observed with noise, which is Lemma 2.1(i). Lemma 2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same reduced-form state. A third example is a desk that each period trades the amount that maximizes $\hat{v}_t u - \frac{1}{2}(\kappa+\lambda)u^2$, a one-period trade-off between its trend forecast $\hat{v}_t$ and quadratic impact and holding costs. Its trade is $u_t = \hat{v}_t/(\kappa+\lambda)$. If the forecast is updated exponentially, $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta x_t$, the trade follows the same recursion, $u_{t+1} = (1-\eta)u_t + w x_t$ with $w = \eta/(\kappa+\lambda)$. This is a static first-order condition, not a dynamic optimum. In these families the memory rate is a known function of a tuning parameter, the desk's signal-to-noise ratio, its regularization or its cost parameters. That is the private information a supervisor would like to recover, although real algorithms need not belong to any of these families. Applied to the pricing error, a desk that cannot separate fundamental news from transitory deviations treats recent deviations as a persistent trend, which is the exponential extrapolation of Barberis et al. (2015, 2018). Nothing in the model says such trading is profitable.

The results use six modelling choices: linearity; price-first timing within a period; exponential memory (needed for the identification result, Theorem 3.3, and the cost results of Section 5, but not for sufficiency); algorithms that do not change their rules when observed (Proposition 6.6 relaxes this for one group); known primitives $(a, \kappa)$; and stated observation contracts. Unknown primitives can only enlarge the set of candidate populations, so the lower bounds remain valid; Corollary S1.2 in the supplement shows that estimating them leaves the upper-bound rate unchanged. Section 4 replaces the primitive $(a, \kappa)$ by an equilibrium.

**The observer.** The cost results are stated for one observer outside the trader groups: a supervisor, an exchange's surveillance unit or a large desk. It knows $(a, \kappa)$, the law of the

observation noise and, for the $K$-group results, $K$. It can *wait*, recording prices driven by natural innovations; *probe*, trading a sequence $v_t$ against the groups from a reset state and recording the response; or *acquire disclosure*, receiving each class's flow. Independent resets are a benchmark that no outside participant can run in a live market. We use them because they make the information content of the data exact. Reset probing is a benchmark for what can be learned, not a proposed policy. Section 5 shows separately that passive observation is no cheaper (Theorem 5.4).

# 3 The kernel is a sufficient statistic

This section states the paper's main result. In a market with an insider, noise traders, a dealer and feedback traders, all following arbitrary causal rules, the feedback traders affect every outcome only through their kernel. The notation for the insider follows Kyle (1985), so in this section $x_t$ is the insider's order, not the pricing error.

**Theorem 3.1** (The kernel is sufficient)**.** *Consider a market with the following participants and rules.*

- *A value $v$ and noise trades $z_t$.*
- *An insider who submits $x_t = X_t(v, \mathcal{H}_{t-1})$.*
- *A dealer who sets $p_t = P_t(\mathcal{H}_{t-1}, y_t)$ from the public history $\mathcal{H}_{t-1}$ and the total order flow $y_t = x_t + z_t + f_t$.*
- *Feedback groups whose memory states update on a public price statistic $s_t = S_t(p_0, \dots, p_t)$ by $m_{i,t+1} = r_i m_{i,t} + s_t$ from $m_{i,0} = 0$, with $r_i = 1 - \eta_i$ and aggregate flow $f_t = \sum_i w_i m_{i,t}$, $w_i = q_i \eta_i$.*

*The rules $X$, $P$ and $S$ are arbitrary causal maps. This covers Kyle, inventory-averse, learning and misspecified dealers, and any insider strategy.*

*(i)* *$f_t = \sum_{m<t} k_m \, s_{t-1-m}$, with $k_m = \sum_i w_i r_i^m$, the moments of the merged measure $M$.*

*(ii)* *Two group configurations produce the same aggregate flow on dates $t \le T$ for every signal sequence $(s_t)$ if and only if $k_m = k'_m$ for all $m < T$. On an unbounded horizon this holds if and only if their merged measures coincide. For a fixed signal rule $S$, the 'if' direction holds for every price path. The 'only if' direction needs a rule that can produce an impulse signal. It fails, for example, for $S \equiv 0$, under which every configuration has zero flow.*

*(iii)* *If $k_m = k'_m$ for $m < T$, then for every strategy profile and every realization of $(v, z)$ the following coincide on $t \le T$: the price and order-flow paths, the insider's trades*

*and realized profit, and the dealer's inventory. The two games therefore have the same strategy sets and the same payoff for every profile. They have the same equilibria, with the same outcomes, under any equilibrium concept defined from these payoffs and the public history.*

*(iv) Group-level quantities are not invariant. A Gauss and a Radau two-point configuration agree on $k_0, k_1, k_2$, so they are outcome-equivalent for $T = 3$, while their largest groups carry weights $\frac{1}{2}$ and $\frac{3}{4}$.*

**In words.** Whatever the insider and the dealer do, two markets whose trader groups have the same merged response produce the same prices, the same order flow and the same insider and dealer outcomes. So the part of the population that prices cannot identify (Theorem 3.3) is worth nothing to anyone who trades on public information. It matters only for actions aimed at a particular group, such as halting it (Proposition 6.1). Parts (i)–(iv) are machine-checked pathwise for every causal insider, dealer and signal rule (`Paper1EquilibriumInvariance.lean`). Supplement S2 gives the proof, the unbounded-horizon statement, the extension to group-specific demand shocks and a numerical illustration.

**Beyond exponential memory.** The proof of part (iii) uses only one property of the feedback traders: their aggregate flow is a fixed linear function of past public signals. So the result does not depend on exponential memory.

**Proposition 3.2** (Any linear population)**.** *In the market of Theorem 3.1, replace the groups by any finite population of feedback traders $j = 1, \dots, J$ whose flows are causal linear functions of the public signal,*

$$f_{j,t} = \sum_{m<t} \kappa_m^{(j)}\, s_{t-1-m}, \qquad \kappa^{(j)} = (\kappa_0^{(j)}, \kappa_1^{(j)}, \dots) \text{ arbitrary real sequences,}$$

*and write $k = \sum_j \kappa^{(j)}$ for the population's kernel.*

*(i) Two populations produce the same aggregate flow on dates $t \le T$ for every signal sequence if and only if their kernels agree for $m < T$.*

*(ii) If their kernels agree for $m < T$, then for every strategy profile and every realization the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory coincide on $t \le T$, and the two games have the same equilibria under any equilibrium concept defined from payoffs and the public history.*

*(iii) Any decomposition of a kernel into individual kernels is consistent with the same outcomes. Prices, flows and all equilibrium outcomes therefore carry no information about the decomposition beyond $k$.*

*Proof.* (i) Aggregate flow is $f_t = \sum_{m<t} k_m s_{t-1-m}$. If the kernels agree for $m < T$, the flows agree on $t \le T$ for every signal. Conversely, the impulse signal $s_j = \mathbf{1}\{j = 0\}$ gives $f_{m+1} = k_m$. (ii) The strong induction in the proof of Theorem 3.1(iii) (Supplement S2) uses only that $f_t$ is the same function of the signals before $t$ in both markets, which is part (i). (iii) follows from (ii), since outcomes depend on the population only through $k$. □

**What the exponential case adds.** For general linear traders, nothing about composition beyond $k$ can be learned from anything that prices determine, because any split of $k$ is observationally equivalent. Exponential memory is the case in which more can be learned, and the rest of the paper prices it. Then $k_m = \int r^m \, dM(r)$ is a moment sequence, and the kernel determines the merged measure $M$ (Theorem 3.1(ii) on an unbounded horizon). A finite record determines only finitely many moments, which is why separating close groups is costly. Proposition 3.2 is machine-checked for arbitrary kernels (`p1_anylinear_iff`, `p1_anylinear_invariance`); the passage to equilibria is on paper, as in Theorem 3.1.

**The analogy with aggregate demand.** The kernel is the response of aggregate feedback demand to past prices. Auclert et al. (2024) show that the analogous object for households, the matrix of intertemporal marginal propensities to consume, is a sufficient statistic for the transmission of fiscal policy. In the same way, $k$ summarizes every heterogeneity among feedback traders that matters for prices. The merged measure is the distribution of individual autoregressive coefficients that governs an aggregate in Granger (1980). Two points distinguish the market version. Sufficiency holds inside a game with an insider and a dealer, not only for the law of an aggregate series. And as part (iv) of Theorem 3.1 shows, it fails for group-level quantities, which matter for targeted supervision (Section 6).

## 3.1 What prices identify

Theorem 3.1 says that populations with the same kernel cannot be told apart by anything that price-based trading determines. The converse is that prices determine the kernel. In the probing experiment of Section 2, the response of prices to trades is a fixed function of $(a, \kappa, k)$, and it determines the merged measure.

**Theorem 3.3** (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state $x_t$.*

*(i)* Equivalence class. *Two economies* $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ *and* $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ *produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0,1]. \tag{8}$$

*Write* $a' = a + \kappa M(\{0\})$*, so that groups with* $\eta = 1$ *(rate* $0$*) act as part of the anchor. For* $a' \neq 0$ *the minimal realization has dimension* $1 + |\mathrm{supp}(M) \cap (0,1)|$*, counting distinct positive rates with nonzero merged weight after signed cancellation; for* $a' = 0$ *the price carries no state of its own and the dimension is* $|\mathrm{supp}(M) \cap (0,1)|$*. A group with* $\eta = 1$ *merges into the anchor* $a$ *without adding a pole.*

*(ii)* Separation complexity under average power. *Consider testing a single group with rate* $\bar{\eta}$ *against two groups with rates* $\bar{\eta} \pm \delta$ *and the same total weight, on a compact family that is uniformly Schur stable with all rates in* $[\epsilon, 1-\epsilon]$ *for some* $\epsilon > 0$*. Observe* $\tilde{x}_t = x_t + \nu_t$ *with* $\nu_t \sim N(0, \sigma_\nu^2)$ *and use deterministic probes with* $\frac{1}{T}\sum_{t<T} v_t^2 \leq P$*. Any test with worst-case error at most* $\alpha \in (0, 1/4)$ *needs*

$$T \;\geq\; \frac{4\sigma_\nu^2(1-2\alpha)^2}{C_{\mathcal{F}}^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{9}$$

*where* $C_{\mathcal{F}}$ *depends only on the family, through* $\epsilon$*, the bound on* $|H|$ *implied by uniform stability, and the total weight* $q$*.*

**In words.** If two trading desks update on prices with the same memory, no amount of price data can tell them apart or say how much each trades: prices reveal only how much trading happens at each memory speed. Desks with different memories can be told apart in principle, and the rest of the paper prices how hard that is.

The memory rates are poles of $G$ and *zeros* of $H$, not closed-loop poles. This has a practical consequence verified in the full working paper: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

*Remark* 3.4 (Scope). The result concerns the stated model and observation contract. Groups sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The $\delta^{-4}$ statement in (ii) is a lower bound on a declared family, with family-dependent constants. A group may choose its rate strategically. The observer then still sees only the merged measure of the rules in use; Proposition 6.6 draws the consequence.

# 4 Sufficiency and identification in equilibrium

The identification results of Section 3 take the price impact $\kappa$ and the persistence $a$ of pricing errors as given. This section derives both from dealers who dislike holding inventory. It shows that the equilibrium price is the closed loop of Section 2 with effective parameters and unchanged memory rates. So the kernel stays the sufficient statistic in equilibrium, and the cost of going beyond it (Section 5) keeps its order.

**An inventory equilibrium.**

- *Fundamental and price.* The fundamental $F_t$ is a public random walk with innovation variance $\sigma_F^2$, and the price is $p_t = F_t + x_t$.
- *Groups.* They trade $u_{i,t+1} = r_i u_{i,t} + w_i x_t$, the paper's exponential memory with $w_i = q_i\eta_i$ and $r_i = 1 - \eta_i$.
- *Exogenous flow.* $\xi_t \sim N(0, \sigma_\xi^2)$, i.i.d., stands for noise trades or the observer's probe. Customer flow is $y_t = \sum_i u_{i,t} + \xi_t$.
- *Dealers.* Competitive dealers absorb the flow and lay off a share $1-\bar{a}$ of their inventory each period, so $H_t = \bar{a}H_{t-1} - y_t$. Each overlapping generation of dealers has CARA risk aversion $A$ and holds $H_t$ to the next date: $H_t = (E_t p_{t+1} - p_t)/(AV)$, with $V = \mathrm{Var}_t(p_{t+1})$.

We study stationary equilibria in which $x_t$ is linear in the dealers' inventory and the groups' next trades. Write $G(D) = \sum_i w_i/(D - r_i) = \int dM(r)/(D - r)$ for the Stieltjes transform of the merged measure.

**Theorem 4.1** (Price impact and persistence in an inventory equilibrium)**.** *Consider stationary linear equilibria of the model above.*

*(i)* Characterization. *The price law*

$$x_t = -\kappa H_t + \kappa \sum_i \frac{u_{i,t+1}}{D - r_i}$$

*with $D \neq 0$ and $D \neq r_i$, is an equilibrium exactly when the induced dynamics are stable and*

$$D = 1 - \kappa\, G(D), \qquad \kappa(D - \bar{a}) = A\, V\, D, \qquad V = \sigma_F^2 + \frac{\kappa^2 \sigma_\xi^2}{D^2}. \tag{10}$$

*An unexpected unit of flow moves the price by $\kappa/D$.*

*(ii)* No feedback. *Then $D = 1$ and the price law is exactly the paper's $x_t = \bar{a}\, x_{t-1} + \kappa\, y_t$. The persistence of pricing errors is the dealers' carry share, $a = \bar{a}$, and the impact $\kappa$ solves*

$\kappa(1-\bar{a}) = A(\sigma_F^2+\kappa^2\sigma_\xi^2)$. *An equilibrium exists if and only if* $\Delta = (1-\bar{a})^2 - 4A^2\sigma_F^2\sigma_\xi^2 \geq 0$. *When* $\Delta > 0$ *there are two,* $\kappa_\pm = \big((1-\bar{a}) \pm \sqrt{\Delta}\big)/(2A\sigma_\xi^2)$*; the low-impact one* $\kappa_- \approx A\sigma_F^2/(1-\bar{a})$ *is the price of dealer inventory risk.*

*(iii)* Invariance. *The coefficients* $(\kappa, D, V)$*, and hence the price process, depend on the groups only through* $G$*, and so only through the merged measure* $M$*. This is Theorem 3.1 in equilibrium form.*

*(iv)* Small feedback. *Let* $\Delta > 0$*, fix* $\delta > 0$ *with* $\max_i r_i < 1-\delta$*, and let* $\epsilon = \sup_{|D-1|\leq\delta}(|G(D)| + |G'(D)|)$ *be small. Then there is a unique equilibrium near* $(\kappa_-, 1)$*. With* $G(1) = \sum_i q_i$*, the aggregate feedback intensity,*

$$D = 1 - \kappa_- G(1) + o(\epsilon), \qquad \kappa = \kappa_- + \frac{\kappa_-^2(\bar{a} + 2A\kappa_-\sigma_\xi^2)}{\sqrt{\Delta}}\, G(1) + o(\epsilon).$$

*Positive feedback raises the impact* $\kappa$*, and it raises the impact of an unexpected flow shock,* $\kappa/D$*, by more.*

**In words.** Dealers set the price impact and the persistence of mispricing. Impact is what they charge for carrying inventory risk; persistence is how slowly they lay that inventory off. Without feedback traders, this equilibrium is exactly the closed loop the paper starts from. Feedback traders enter only through their merged measure, as Theorem 3.1 requires. Positive feedback raises price impact, and dealers price their predictable future trades in advance (the terms in $u_{i,t+1}$). Feedback trading also makes liquidity fragile, not only more expensive: beyond a threshold in dealer risk aversion that falls with the feedback intensity, no equilibrium exists (Theorem S2.2).

**Machine check.** See Supplement S20.

**The identification results hold in equilibrium.** With feedback, dealers price the groups' predictable trades in advance, so the response of prices to flow is no longer the reduced form of Section 2 with the primitive $(a, \kappa)$. The next theorem shows that it is still that closed loop, exactly and path by path, with new parameters and the same memory rates.

**Theorem 4.2** (The equilibrium is the paper's closed loop)**.** *Take an equilibrium of Theorem 4.1 with* $D \neq \bar{a}$*, and set*

$$a^* = \frac{\bar{a}}{D}, \qquad \kappa^* = \frac{\kappa}{D}, \qquad \tilde{w}_i = \frac{D-\bar{a}}{D-r_i}\, w_i.$$

*(i)* Reduced form, pathwise. *Let* $\tilde{u}_{i,t} = \frac{D-\bar{a}}{D-r_i} u_{i,t}$. *For every realization, every initial state and every* $t$,

$$x_t = a^* x_{t-1} + \kappa^* \Big( \xi_t + \sum_i \tilde{u}_{i,t} \Big), \qquad \tilde{u}_{i,t+1} = r_i \tilde{u}_{i,t} + \tilde{w}_i x_t.$$

*So the price follows the closed loop* (1)–(2) *with persistence* $a^*$, *impact* $\kappa^*$, *exogenous flow* $\xi$ *in place of the probe, and groups with the same memory rates and merged weights* $\tilde{w}_i$. *Its transfer function from flow to price is* (4) *with* $(a^*, \kappa^*, \tilde{w})$. *The equilibrium is stable if and only if this closed loop is.*

*(ii)* Zeros. *For* $\bar{a} > 0$, *distinct rates in* $(0, 1)$ *and nonzero weights, the memory rates are the transmission zeros of the equilibrium price response, and none cancels (Theorem 5.3(i) with the effective parameters).*

*(iii)* Boundary. *The effective merged measure is* $\tilde{M} = h\, M$, *where* $h(r) = (D - \bar{a})/(D - r)$ *is positive. The effective parameters determine the equilibrium ones:* $D = 1 - \kappa^* \tilde{M}([0, 1])/(1 - a^*)$, $\bar{a} = a^* D$ *and* $\kappa = \kappa^* D$. *So the merged measure remains exactly what equilibrium prices identify (Theorem 3.3).*

*(iv)* The rates survive. *Let* $(a^*, \kappa^*, W, r_c, \Delta\eta)$ *satisfy the hypotheses of Theorem 5.1 with* $(a, \kappa) = (a^*, \kappa^*)$. *Then the two* $K$*-group economies of that theorem are the price responses of two inventory equilibria with identical primitives* $(A, \sigma_F^2, \sigma_\xi^2, \bar{a})$ *and identical* $(\kappa, D)$*:*

- $D = 1 - \kappa^* W/(1 - a^*)$, *which exceeds every rate;*
- $\bar{a} = a^* D$ *and* $\kappa = \kappa^* D$;
- *any* $\sigma_F^2, \sigma_\xi^2 > 0$, *with* $A = \kappa^* D(1 - a^*)/(\sigma_F^2 + \kappa^{*2} \sigma_\xi^2)$;
- *group weights* $w_i = \tilde{w}_i (D - r_i)/(D - \bar{a}) > 0$.

*Hence an observer of equilibrium prices who knows the primitives needs* $N \geq c\, \sigma^2 (\Delta\eta)^{-(4K-2)}$ *experiments. Theorem 5.2 applies to the effective parameters, and the true weights follow from the effective ones by a Lipschitz map. So* $N^* \asymp \sigma^2 (\Delta\eta)^{-(4K-2)}$ *in equilibrium.*

**In words.** Dealers' anticipation does not hide the groups and does not move them. It replaces the persistence and impact by $\bar{a}/D$ and $\kappa/D$, and it rescales each group's weight by $(D - \bar{a})/(D - r_i)$. Because that factor rises with $r_i$, slow-memory groups weigh more in equilibrium prices. The memory rates are untouched.

- Every result of Sections 3 and 5 that concerns the map from order flow to prices therefore holds in equilibrium: identification, sample complexity, passive observation, and probing costs computed from the price path. Its constants are evaluated at the effective parameters.
- The lower-bound pair is not an artefact of the reduced form. Two economies with the same dealers, the same risk aversion and the same impact differ only in their groups, and they are as hard to tell apart as Theorem 5.1 says.
- Part (i), the realization in part (iv) and the effective group recursion are machine-checked (`Paper1EquilibriumReducedForm.lean`).
- One assumption is new: the observer's probe is part of the exogenous flow that dealers price as noise. Dealers who recognized a probe would price it differently.

**Further equilibrium results.** Four results in Supplement S2 extend this section. None is needed for the main argument.

- *A strategic insider* with short-lived information (Theorem S2.3). The equilibrium is characterized in closed form and nests Kyle's one-period values. Its coefficients depend on the groups only through $G$, and the observed price is the closed loop driven through a known invertible moving-average filter. So the identification boundary and the $(\Delta\eta)^{-(4K-2)}$ rate hold with an insider present. Positive feedback raises total impact, and the insider then trades less and prices reveal less of her news.
- *Fragility* (Theorem S2.2). With positive feedback, a positive-impact equilibrium exists only if dealer risk aversion lies below a threshold. For strong feedback the threshold is inversely proportional to the feedback intensity $G(1)$. With several groups the bound depends on the population only through $G(1)$, again an aggregate.
- *Rents* (Theorem S2.4). A dealer that misjudges the aggregate response by $\varepsilon$ hands an insider with a committed schedule a rent of exact order $\varepsilon$, or charges it one. Public data reveal the aggregate response at the ordinary rate. So the information that protects price formation is cheap to learn.
- *A targeted halt in equilibrium* (Theorem S2.5). Two markets that agree in equilibrium to within the cube of the memory spacing can differ at order one after their largest group is halted.

Disclosure keeps its value in equilibrium too (Corollary S2.1).

# 5 The price of composition

Prices identify the kernel, and with exponential memory the kernel determines the merged measure. A finite record, however, pins down only finitely many moments of $M$, each with noise. This section asks how much data it takes to recover the groups. The answer is the known rate for finite mixtures. What the market adds is that the memories must first be extracted from a feedback loop, and that waiting is no cheaper than trading.

## 5.1 Many unknown groups

**Theorem 5.1** ($K$-group two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0,1)$, total merged weight $W > 0$ and $0 < \Delta\eta < \min(r_c, 1 - r_c)$, so that all rates lie in $(0,1)$, and assume $a + \kappa W/(1 - r_c - \Delta\eta) < 1$, so that every candidate economy is Schur stable. Each reset experiment observes the noisy taps $h_0, \dots, h_{T-1}$ with $T \geq 2K + 1$ ($T = \infty$ allowed), so that the first tap where the two economies below differ, $h_{2K}$, is observed; the bound below uses the full response norm, which dominates any finite prefix. There exist two $K$-group economies with rates in $r_c + \Delta\eta[-1, 1]$, positive weights, and merged measures whose moments of order $0, \dots, 2K-2$ coincide while the moment of order $2K-1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from $N$ independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

*The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.*

The two losses have different targets. The weights of the two economies differ by a constant that does not shrink with $\Delta\eta$, so the first statement is about weights at fixed accuracy, which Theorem 5.2 matches; the Wasserstein statement is about the memory distribution at accuracy proportional to $\Delta\eta$, which the full working paper matches with an upper bound.

The construction uses the $K$-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have $K$ atoms and integrate polynomials of degree $2K - 2$ exactly. For $K = 2$ it reproduces the order of Theorem S4.1. Numerically, their low moments agree to machine precision and their moment of order $2K - 1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with $K$ on top of the exponent. The two rules also differ in their largest weight for every $K$ we checked: for

$K = 2, \dots, 200$ the largest Gauss–Radau weight exceeds the largest Gauss weight, by 0.25 at $K = 2$ and by about $0.79/K^2$ for large $K$ (`equilibrium/gauss_radau_maxweight.py`). So the argument given for $K = 2$ in Proposition 6.1 applies to each such $K$. A proof of this comparison for every $K$ is open. The supervisory result needs less: Lemma S5.3 proves that, for every $K$, some two rules of this kind have different largest weights.

**What counts as an experiment.** In Theorem 5.1 an experiment is a unit impulse from the reset state whose response is observed with noise of variance $\sigma^2$ per tap. The exponent is the same for any probe design of bounded energy. For any probe $v$ the mean difference between the two economies is at most $\sup_\omega |\Delta\hat{H}(e^{i\omega})|\,\|v\|_2$, and $\sup_\omega |\Delta\hat{H}| = O((\Delta\eta)^{2K-1})$ because $\Delta\hat{H} = z^{-1}\hat{H}_A\hat{H}_B\,\Delta G$ with both responses bounded on the circle. Experiments with probe energy $E$ each therefore need $NE \gtrsim \sigma^2(\Delta\eta)^{-(4K-2)}$: a larger probe buys the same information with fewer experiments, and the account-level analysis of the full working paper prices the capital that a larger probe requires. The bound does not cover information obtained without trading against the groups, such as class records (Section 6).

**Theorem 5.2** (Matching upper bound for every $K$)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i\neq j} |s_i - s_j| \geq \gamma > 0$, and merged weights $w_i \in [\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of $K$-group economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and $N$ independent reset experiments, each observing at least the taps $h_0, \dots, h_{2K}$ with Gaussian noise of variance $\sigma^2$. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2}(\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(4K/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \;\leq\; C\sqrt{\log(4K/\beta)}\;\epsilon_N$$

*with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence. On any such class that contains the Gauss and Gauss–Radau pair of Theorem 5.1 (a rich class), together with that theorem the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$.*

**Smaller classes and constants.** On smaller classes the upper bound need not be tight: if the rates are known, for example, two weights need only order $\sigma^2(\Delta\eta)^{-2}$ experiments. For $K$ known clustered rates the requirement is at least of order $\sigma^2(\Delta\eta)^{-2(K-1)}$. A weight perturbation along the $(K-1)$-th finite difference preserves the moments of order $0, \dots, K-2$ and changes the moment of order $K-1$ by order $(\Delta\eta)^{K-1}$. For $K = 3$, the direction $(1, -2, 1)$

on rates $r_c - \Delta\eta, r_c, r_c + \Delta\eta$ changes the second moment by $2(\Delta\eta)^2$. This is still far below the unknown-rate order $(\Delta\eta)^{-(4K-2)}$. The constants $c_0$ and $C$ grow quickly with $K$, through the conditioning of the moment map; the theorem gives the order in $\Delta\eta$ for fixed $K$, not a usable constant for large $K$ (in the full working paper our double-precision Prony implementation already fails at $K = 4$ on the disclosure grid).

**In words.** With $K$ unknown groups whose memories lie within $\Delta\eta$ of each other, the number of experiments needed grows like $(\Delta\eta)^{-(4K-2)}$. For two groups, halving the spread multiplies the data needed by 64; for three groups, by 1,024. Theorem 5.1 shows that no method does better, and this theorem shows that Prony's method achieves it.

## 5.2 Passive data

**Theorem 5.3** (Passive identification: minimum phase)**.** *Consider a stable economy with $0 < a < 1$, distinct memory rates $0 < r_i < 1$ and nonzero merged weights.*

*(i) The zeros of $H$ are the memory rates $r_i$ together with a structural zero at $z = 0$; none cancels against the denominator, and all lie in the open unit disk, so $H$ is minimum phase. The degenerate anchor ($a = 0$) and zero-rate cases require the cancellations described in the proof of Theorem 3.3.*

*(ii) Let the price be driven by a single innovation sequence with variance $\sigma_\xi^2 > 0$, possibly unknown, so that its spectrum is $f(\omega) = \sigma_\xi^2 |H(e^{i\omega})|^2/(2\pi)$. Within the class of (i), $f$ determines $\sigma_\xi^2 \kappa^2$ and the normalized response $H/h_0$ (recall $h_0 = \kappa$); hence it determines the memory rates. If in addition $\kappa$ (or $\sigma_\xi^2$) and $a$ are known, it determines $H$ and, by Theorem 3.3, the merged measure.*

The result holds for signed weights (mean-reverting groups): in 1,733 stable random economies with weights of both signs, the nonstructural zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the groups' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

**Theorem 5.4** (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of i.i.d. Gaussian innovations $\xi_t \sim N(0, \sigma^2)$, $\sigma^2 > 0$ known, without observation noise, for $\theta$ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

*(i)* *For a single group at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.*

*(ii)* *For the moment-matched two-group economies of Theorem S4.1, $D = \Theta(\tau^2(\Delta\eta)^6)$.*

*(iii)* *For every $K \geq 2$, for the $K$-group Gauss and Gauss–Radau economies of Theorem 5.1, $D = \Theta((\Delta\eta)^{4K-2})$.*

*Hence any test based on $T$ passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, uniformly as the separation shrinks: passive data cost at least the same polynomial order as active probing for every number of groups. For each fixed pair, at fixed type-I error the likelihood-ratio test's type-II error decays at exponential rate $D$ (Stein's lemma), and both errors decay at the Chernoff-information rate, which is of the same order in the separation. So the orders are the right ones. We do not claim that this attainment is uniform as the separation shrinks. All constants depend only on $K$, the stability margin of the family, the rate interval $[\eta_{\min}, \eta_{\max}]$, the total weight and the moment-matched construction; they do not depend on $\delta$, $\Delta\eta$ or $T$. In part (ii) $\tau$ is held fixed as $\Delta\eta \to 0$. The testing conclusions use a finite-record bound $D_{\mathrm{KL}}(P_1^T \| P_0^T) \leq C\,T\,\varepsilon^2$, with $\varepsilon$ the leading moment gap ($\delta^2$, $\tau(\Delta\eta)^3$, $(\Delta\eta)^{2K-1}$); the lower bounds on $D$ and the fixed-pair error exponents are separate statements.*

**In words.** Watching the market without trading does not escape the barrier. A passive observer needs at least the same polynomial order of observations as a prober needs experiments; it pays in time rather than in trading losses. Prices recorded with extra measurement noise that is independent of the price are a garbling of the noiseless record, so by the data-processing inequality they carry less information, and the lower bound holds a fortiori.

**Machine checks.** The machine checks of Theorems 5.1–5.4 are described in Supplement S20.

**In executed quantity and money.** For a *known* pair (one group at rate $\eta$ against two at $\eta \pm \delta$ with the same total weight), the information needed can be stated in trades. The executed quantity a prober needs is of exact order $\delta^{-2}$: the two price responses first differ at lag two, by $-\kappa^2 q \delta^2$, and a single pulse of size $2\sigma\Phi^{-1}(1-\alpha)/(\kappa^2 q\delta^2)$ attains error $\alpha$. The capital a counterparty must post to fill such probes is of order $\delta^{-4}$, because the adverse mark of a probe grows with the square of its size. Halving the difference between two memories therefore quadruples the trading needed and multiplies the capital by sixteen (Theorems S3.1 and S3.3 and Lemma S3.5, Supplement S3).

**Near instability.** Theorem 5.4 is stated on a family that stays uniformly stable. One might hope that the barrier lifts as the market approaches instability, since the loop then amplifies every difference. It does not. What becomes visible is the distance to instability, not the composition.

**Proposition 5.5** (The barrier does not lift near instability)**.** *Fix $K \geq 2$ and the $K$-group Gauss and Gauss–Radau economies $A$ and $B$ of Theorem 5.1, with positive weights, rates in $(0, 1)$ and $\kappa > 0$. For a margin $\epsilon > 0$, choose persistences $a_A(\epsilon)$ and $a_B(\epsilon)$ so that both economies have feedback budget $S = 1 - \epsilon$, and suppose $a_A(\epsilon), a_B(\epsilon) > 0$ for $\epsilon \leq \epsilon_0$. Let $D_\epsilon(B\|A)$ be the Kullback–Leibler divergence rate between the stationary price processes driven by i.i.d. $N(0, \sigma^2)$ innovations. There is $C$, not depending on $\epsilon$ or $\Delta\eta$, such that*

$$D_\epsilon(B\|A) \;\leq\; C\,(\Delta\eta)^{4K-2} \qquad \text{for all } 0 < \epsilon \leq \epsilon_0 \text{ and all small } \Delta\eta.$$

**In words.** Even as the market approaches a crash, passive prices reveal no more about who is trading than they did in calm conditions, per observation. They reveal more and more about how close the market is to instability: if the two economies share the persistence $a$ instead of the budget, their budgets differ by order $(\Delta\eta)^{2K-1}$, and numerically the divergence rate then grows like $\epsilon^{-1}$. The proof is in Supplement S5.2. Two limits apply. The statement is about the divergence rate per observation; near instability prices are strongly dependent, and a finite-sample bound that is uniform in $\epsilon$ is not proved, although numerically the exact information in $n$ observations stays within 3% below $n$ times the rate down to $\epsilon = 10^{-3}$ (Supplement S5.2). And probing does gain near instability: the squared response difference grows like $\epsilon^{-1}$ (numerical, `research_extension/fragility_reveals`), but probing a market at the edge of instability is not a policy. Figure 1 shows both statements.

**Monetary value of algorithm tags.** The sample complexity gap converts directly into a dollar value of mandatory class-tagged disclosure. Let $c > 0$ be the per-period opportunity cost of market surveillance or active supervisory probing (e.g., $c = \sigma_w^2/\bar{\kappa}$ in dollar terms). To attain target statistical power $1-\alpha$, a supervisor without tags must incur passive surveillance expenditure $C_{\text{pass}} = cN_p$, whereas under mandatory class disclosure the expenditure is $C_{\text{tag}} = cN_t$. The monetary Value of Disclosure (VOD) is the avoided surveillance cost:

$$\text{VOD}(K, \Delta\eta, c, N_t) = C_{\text{pass}} - C_{\text{tag}} = c \cdot N_t \left[(\Delta\eta)^{-(4K-2)} - 1\right]. \tag{11}$$

By Theorem 6.3, this avoided expenditure is non-negative, scales linearly in surveillance cost $c$ and sample size $N_t$, and grows without bound as $\Delta\eta \to 0$ (`p1_disclosure_monetary_value`,

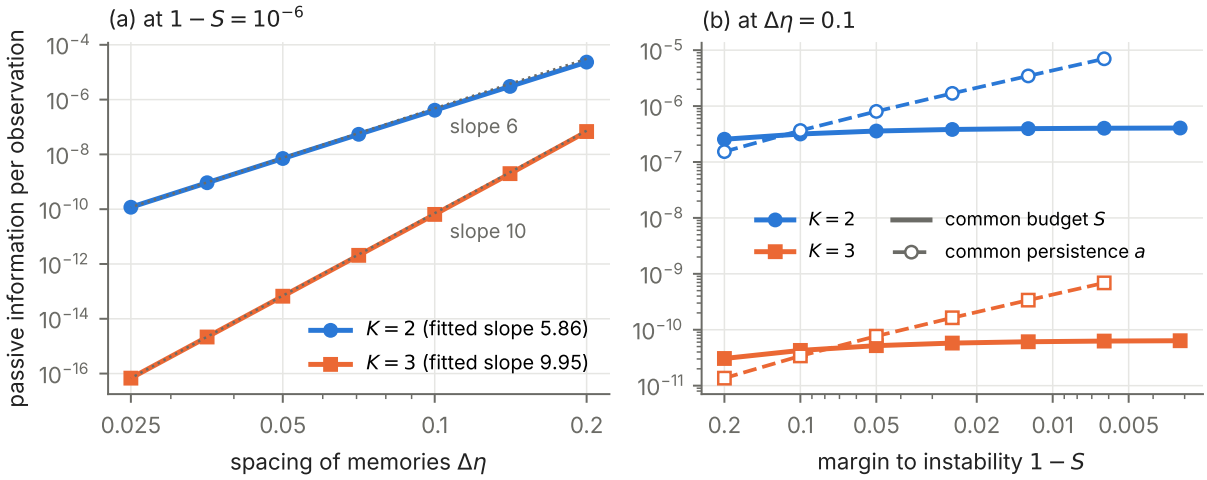



Figure 1: The passive barrier near instability. Both panels show the Gaussian Kullback–Leibler information per observation between the stationary price processes of two populations whose merged measures are the $K$-node Gauss and Gauss–Radau rules on memories spread over $\Delta\eta$ around 0.5 (total weight 0.3, impact $\kappa = 0.5$). (a) At a common feedback budget one millionth below instability, the information falls like $(\Delta\eta)^{4K-2}$ (dotted lines). (b) Held at a common budget (solid), the information stays bounded as the margin $1 - S$ shrinks; held at a common persistence (dashed), it grows without bound, because the budgets then differ. Computed from the closed-form spectra; scripts in `research_extension/fragility_reveals/`.

`p1_disclosure_vod_as_cost_difference`, `p1_disclosure_value_dominates_audit_cost`). In financial markets with $K = 2$ groups and group separation $\Delta\eta = 0.1$, the sample complexity ratio is $(\Delta\eta)^{-6} = 10^6$ (`p1_disclosure_ratio_K2_delta_eta01`). At a modest surveillance cost of $c = \$10$ per market-day with $N_t = 100$ tagged records required, passive surveillance would demand $\$10 \times 10^8 = \$1{,}000{,}000{,}000$ in observation expenditure, yielding VOD $\approx \$10^9$ in averted surveillance waste (`p1_disclosure_vod_example`, `p1_disclosure_cost_ratio_K2`). Any fixed compliance or auditing cost incurred to implement algorithm tagging is therefore dominated by several orders of magnitude.

# 6 Who needs composition, and what disclosure buys

By Theorem 3.1, composition is worth nothing to anyone trading on public information. It can still matter to a supervisor whose action is aimed at one group. This section gives a decision in which it does and the disclosure that removes the cost of learning it. It ends with a bound on what price-based targeting can achieve when a group can hide.

**A decision in which attribution matters.** A supervisor must keep the feedback budget $S = a+\kappa \sum_i q_i$ below a prudential ceiling $\bar{s}$. It can either cut all activity uniformly or halt the largest verified class. The target is the feedback budget, not the spectral radius of the state matrix: in an already stable positive economy, imposing $S \le \bar{s} < 1$ is a tighter prudential constraint, not a restoration of stability.

**Proposition 6.1** (Supervisory separation: market-wide versus targeted intervention)**.** *Assume the positive stable model, known $a, \kappa$, and $a < \bar{s} < 1$. Write $q_i = w_i/(1 - r_i)$ and $R = S - \kappa \max_i q_i$ for the post-halt feedback budget.*

*(i) A uniform scale $\theta$ meets the ceiling exactly when $a + \theta(S - a) \le \bar{s}$. Its largest feasible value is $\theta^* = \min\{1, (\bar{s} - a)/(S - a)\}$. The identity $S = 1 - \kappa/\sum_j h_j$ is algebraic; estimating this infinite sum requires a justified tail bound and an observation/noise model. It alone establishes no unconditional parametric learning rate.*

*(ii) For $K = 2$, the Gauss and Gauss–Radau pair of Theorem 5.1 has limiting largest normalized weights $1/2$ and $3/4$. Thus $|S^G - S^R| = O((\Delta\eta)^3)$ and $q^R_{\max} - q^G_{\max} = W/(4(1 - r_c)) + O(\Delta\eta)$. For sufficiently small spacing, a nonempty interval of ceilings separates their post-halt decisions. For $N$ independent reset records with common Gaussian noise $\sigma^2 I$, both decision errors at most $1/4$ require*

$$N \ge (2\Phi^{-1}(0.75)\sigma/\|h^G - h^R\|_2)^2 \ge c\sigma^2(\Delta\eta)^{-6}.$$

*The record horizon and the response norm must be the same on both sides.*

*(iii) Tagged-flow estimation under Theorem 6.3, a rate bound away from one, and a fixed nonzero decision margin gives a bounded sample requirement as the spacing shrinks. The target class must be mapped to an executable, verified class label; a largest latent atom alone supplies no such identity.*

The proof is in Supplement S5. Proposition S5.4 there extends part (ii) to every $K$, with exponent $4K - 2$.

**In words.** A uniform cut needs only the kernel, through $S = 1-\kappa/\sum_j h_j$, so public prices suffice for it. A halt of the largest group needs that group's weight, which the kernel does not pin down. The two-group populations of Theorem 5.1 (the Gauss and Gauss–Radau pair with $K = 2$) have feedback budgets within $O((\Delta\eta)^3)$ of each other, but largest groups that differ at order one. So a range of ceilings exists at which halting the largest group meets the ceiling in one population and not in the other. Prices decide between them only with

order $(\Delta\eta)^{-6}$ records. The same holds for any number of groups $K$, with order $(\Delta\eta)^{-(4K-2)}$ records (Proposition S5.4).

## 6.1 What class-tagged disclosure buys

**Proposition 6.2** (Tagged flow removes the barrier)**.** *Let the price $x_t$ be the causal stationary solution of the model driven by i.i.d. centred innovations of variance $\sigma_\xi^2 \in (0,\infty)$ in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose group $i$ has merged weight $w_i > 0$ and the observer receives prices and each group's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. centred measurement noise of finite variance $\sigma_f^2$, independent of the whole innovation sequence and hence of the whole price path. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2 \big[ (1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top) \big], \quad (12)$$

*where $A_m = \sum_{s \geq 0} r_i^s \gamma_x(s+m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ and $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. The Jacobian determinant is*

$$\det \Sigma_i = w_i \sum_{\ell,m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \; > \; 0. \quad (13)$$

*$V_i$ depends only on group $i$'s own $(w_i, r_i)$, $\sigma_f^2$ and the price autocovariances. Two conclusions are distinct: (a) for each admissible economy, $\hat{\beta}_i$ has the stated limiting normal law; (b) on a uniformly stable family with stationary price variance at least $\gamma_{\min} > 0$ and merged weights at least $w_{\min} > 0$, the limiting covariance $V_i$ is bounded by a constant that does not depend on $|r_i - r_j|$. Statement (b) bounds the limit; it is not a central limit theorem that holds uniformly over the family.*

**Machine check.** Three pieces are machine-checked: the finite-sample reset-experiment version, with an error bound that does not depend on the spacing; the lower bound on the determinant of the population instrument matrix that does not depend on the spacing (`p1_tagged_iv_determinant_bound`); and the exact decomposition of the IV score into reporting-error terms with coefficients fixed by the price path plus two boundary terms (`p1_iv_boundary_decomposition`). The stationary law of large numbers and the central limit theorem behind (a) are not machine-checked.

**In words.** If each group's trading is reported separately, each group becomes an ordinary regression. Its precision depends on the group's own rate and on the price dynamics, but it has no singularity as the groups' rates approach each other.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about $w_i$ does not involve the other groups' rates; the closed form (13) matches the simulated Jacobian (0.1330 against 0.1322 in a two-group economy with $4 \times 10^5$ periods). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between $5.8 \times 10^{-4}$ and $6.6 \times 10^{-4}$). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem. Proposition 6.2 says that tagging removes the barrier; the next result says by how much.

**Theorem 6.3** (Value of class-tagged disclosure)**.** *Take a rich class of Theorem 5.2: rates $r_c + \Delta\eta\, s$ with $s$ in a compact set of $\gamma$-separated ordered points that has nonempty interior and contains the Gauss and Gauss–Radau pair of Theorem 5.1, and merged weights in a nondegenerate interval $[\omega_{\min}, \omega_{\max}]$, $0 < \omega_{\min} < \omega_{\max}$. In each reset experiment the observer applies a unit impulse and records the price response with independent $N(0, \sigma^2)$ noise per tap; under tagging it also records each class's flow response $f_{i,t} = w_i \sum_{s \le t} r_i^{t-s} h_s$ with independent $N(0, \sigma_f^2)$ noise per tap. Let $N_p$ and $N_t$ be the numbers of experiments needed, without and with tagging, for a fixed accuracy and confidence. There are constants $0 < c \le C < \infty$, depending on $(\sigma, \sigma_f)$ and the class but not on $\Delta\eta$, such that*

*(i) to recover the weights to a fixed accuracy smaller than half the weight difference between the Gauss and Gauss–Radau economies (so that recovering the weights decides between them), $c \le N_t \le C$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-2)}$;*

*(ii) to recover the rates to accuracy $c_0\Delta\eta$, for a fixed $c_0 > 0$ smaller than the radius of a ball contained in the class of rescaled nodes (so that no fixed guess meets the target), $c(\Delta\eta)^{-2} \le N_t \le C(\Delta\eta)^{-2}$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-4)}$.*

**In words.** This is the value of disclosure measured in data. Tagging replaces a requirement that explodes as the groups become similar with one that does not grow at all (for the weights) or grows only like $(\Delta\eta)^{-2}$ (for the rates).

Proposition 6.2 and Theorem 6.3 use different observation arrangements: the proposition concerns passive, stationary data with prices observed without noise, where the price is a valid instrument; the theorem concerns reset experiments with noisy prices. With noisy prices, regressing flow on the observed price is an errors-in-variables problem, which is why the theorem works from the known impulse instead.

## 6.2 Partial disclosure

Actual regimes tag some traders and not others: Japan registers high-speed traders, India identifies algorithmic orders, and everyone else stays anonymous. Theorem 6.3 prices full tagging. The next result prices tagging some of the groups.

**Theorem 6.4** (Partial disclosure)**.** *In the reset experiments of Theorem 6.3, let the flows of $m$ of the $K$ groups be recorded (tagged), and write $K' = K - m$ for the number of untagged groups.*

*(i)* Lower bound. *If $K' \geq 2$, there are two $K$-group economies with rates in $r_c + \Delta\eta[-1, 1]$ and positive weights, with identical tagged groups and with untagged groups forming the $K'$-node Gauss and Gauss–Radau pair of Theorem 5.1, such that telling them apart from $N$ reset experiments, using the price records and the tagged flow records, with error probability below $1/4$ needs $N \geq c\,(\Delta\eta)^{-(4K'-2)}$.*

*(ii)* Upper bound. *Let the tagged groups satisfy the conditions of Theorem 6.3, and let the untagged groups range over a rich class of Theorem 5.2 with $K' \geq 2$ nodes. Estimate each tagged group from its flow records, subtract its contribution from the estimated kernel, and apply Prony's method with $K'$ nodes to the residual. This recovers the weights of all $K$ groups to any fixed accuracy from $N = C(\Delta\eta)^{-(4K'-2)}$ experiments.*

*(iii)* One untagged group. *If $K' \leq 1$, the weights of all $K$ groups are recovered from a number of experiments that does not depend on $\Delta\eta$.*

*The constants depend on $(\sigma, \sigma_f)$, the class and the tagged groups, not on $\Delta\eta$. So the cost of composition is that of the untagged groups alone: $N^* \asymp (\Delta\eta)^{-(4K'-2)}$ for $K' \geq 2$.*

*Proof.* (i) The two economies share the tagged groups, so their kernels differ only through the untagged pair, whose moments agree up to order $2K' - 2$ (Theorem 5.1); the kernel gap is therefore of order $(\Delta\eta)^{2K'-1}$ at every lag. The price impulse responses inherit this order through the closed-loop recursion, as in the proof of Theorem 5.1, with constants that depend on the tagged groups only through the closed-loop stability margin. A tagged group's flow response, $f_{i,t} = w_i \sum_{s \leq t} r_i^{t-s} h_s$, is the same linear map of the price response in both economies, so its mean difference is also of order $(\Delta\eta)^{2K'-1}$. With Gaussian noise the divergence of one experiment's full record is at most $C(\Delta\eta)^{4K'-2}$, and Le Cam's two-point lemma gives the bound. (ii) As in the proof of Theorem 6.3, the averaged price responses and least squares on each tagged class's flows estimate the tagged weights and rates with error $O_P(N^{-1/2})$, uniformly on the class and without dependence on $\Delta\eta$ or on the other groups. The kernel values $k_0, \ldots, k_{2K'-1}$ are smooth functions of the taps (proof of

Theorem 5.2), so the residual kernel $k_j - \sum_{\text{tagged}} \hat{w}_i \hat{r}_i^j$ estimates the untagged groups' kernel with error $O_P(N^{-1/2})$. Prony's method on these $2K'$ values is the estimator of Theorem 5.2 with $K'$ nodes, whose error is the input error times order $(\Delta\eta)^{-(2K'-1)}$ (the rescaling of the centred moments). So the untagged weights have error $O_P(N^{-1/2}(\Delta\eta)^{-(2K'-1)})$, and $N = C(\Delta\eta)^{-(4K'-2)}$ suffices; the tagged weights are recovered at the faster rate. (iii) With one untagged group of weight $v$ and rate $s$, $k_0 = \sum_{\text{tagged}} w_i + v$ and $k_1 = \sum_{\text{tagged}} w_i r_i + vs$, so $v$ and $s$ are smooth functions of $k_0$, $k_1$ and the tagged estimates, all estimated with error $O_P(N^{-1/2})$ independent of $\Delta\eta$. With none, this is Theorem 6.3(i). □

**The value of each tag.** Tagging one more group lowers the exponent by four while two or more groups stay untagged: the data needed fall by a factor of order $(\Delta\eta)^{-4}$. The tag that leaves a single group untagged is worth the most, a factor of order $(\Delta\eta)^{-6}$, because one untagged group needs no separating at all. The last tag is worth nothing for the weights. So a supervisor never needs to tag everyone: tagging all but one memory class gives the full composition at a cost that does not depend on how close the memories are. What matters is how many distinct memory classes remain untagged, not how many traders. A regime that tags registered high-speed traders and leaves everyone else anonymous gets the full benefit only if everyone else behaves, in memory, like a single class.

**Numerical check and machine check.** For $K = 3$ with no tag and one tag, and $K = 4$ with two tags, the exact divergence per experiment of the hard pair falls with $\Delta\eta$ at fitted slopes 10.05, 6.02 and 5.96, against 10, 6 and 6 (`research_extension/partial_disclosure/partial_disclosure_check.py`). Part (i) is machine-checked on the joint records of prices and tagged flows, for records of any fixed length $T$ and a common noise variance $\sigma^2$ on prices and flows (with $\sigma_f \neq \sigma$, replace $\sigma$ by $\min(\sigma, \sigma_f)$): for any two economies that share their tagged groups and whose untagged node moments agree below order $n$, any test satisfies $4\sigma^2(1-\alpha_0-\beta_0)^2 \le N(T+MT)\big(2W2^T(\Delta\eta)^n C_T(1+W_tT)\big)^2$ (`p1_partial_lower_bound_matched`, `Paper1PartialDisclosureLower.lean`); with the matched $K'$-node pair of Theorem 5.1, $n = 2K'-1$. The algebraic steps of (ii) and (iii) are machine-checked too (`Paper1PartialDisclosure.lean`), and so is the deterministic core of (ii): if the kernel taps are estimated within $e_k$ and the tagged weights and rates within $e_w$ and $e_r$, an ordered $K'$-node representation matching the residual moments exists, is unique and recovers every untagged weight within $L\,2^{2K'}\eta(\Delta\eta)^{-(2K'-1)}$, with $\eta = e_k + m(e_w + W_t\,2K'\,e_r)$ (`p1_partial_upper_deterministic`, `Paper1PartialDisclosureUpper.lean`). The $O_P(N^{-1/2})$ rates of the inputs are on paper, and the whole estimator is checked by Monte Carlo: subtract-then-Prony gives weight errors growing like $(\Delta\eta)^{-(2K'-1)}$, fitted

slopes $-5.07$, $-3.02$ and $0.36$ for $K' = 3, 2, 1$ untagged groups of three, against $-5$, $-3$ and $0$ (`research_extension/theorem_checks/mc_rates.py`).

## 6.3 Pooling markets

Algorithms trade in many markets at once. A trend follower's lookback is a property of the algorithm, the same in every futures contract it trades, while how much it trades in each market varies. A supervisor who sees several markets therefore sees several kernels built from the same memory rates with different weights. The next result says what that is worth.

Take $D$ markets, each with known persistence and impact and its own independent noise, and the same $K$ algorithms with memory rates $r_i = r_c + \Delta\eta\, s_i$ in every market and merged weights $W_{ij} > 0$ in market $j$. Every market is Schur stable for every configuration considered, $a_j + \kappa_j \sum_i W_{ij}/(1 - r_c - \Delta\eta) < 1$. In each market the supervisor runs $N$ reset experiments as in Theorem 5.2. Let $q^* = \lceil K/D \rceil$ and $l^* = K - 1 + q^*$.

**Theorem 6.5** (Pooling markets)**.** *(i)* Lower bound. *Near every configuration $(s, W)$ with distinct nodes at which the moment map below order $l^*$ has locally constant rank (an open dense set), there are two configurations at a fixed distance in $(s, W)$, with distinct nodes and positive weights, whose kernels agree in every market up to the moments of order $l^* - 1$. Telling them apart with error probability below $1/4$ needs $N \geq c\,(\Delta\eta)^{-2l^*}$.*

*(ii)* Upper bound. *At configurations where the rescaled moment map up to order $l^*$ has an injective differential (condition (R)), a minimum-distance estimator restricted to a neighbourhood recovers the rescaled nodes and the weights with error $O_P(N^{-1/2}(\Delta\eta)^{-l^*})$, so $N = C(\Delta\eta)^{-2l^*}$ experiments suffice locally.*

*(iii)* The order is sharp. *Condition (R) never holds below order $l^*$, and it holds at generic configurations for every $K \leq 8$ and $D \leq K$.*

*(iv)* Globally. *Take a compact class of configurations with nonempty interior, separated nodes and weights bounded below. Let $l_c$ be the least order at which the moment map is injective on the class up to relabelling, and $l_g \geq l_c$ the least order at which it is, in addition, an immersion on the class. Over that class the data needed are at least of order $(\Delta\eta)^{-2l_c}$ and at most of order $(\Delta\eta)^{-2l_g}$, with $l^* \leq l_c \leq l_g \leq 2K - 1$; if condition (R) holds at order $l^*$ everywhere on the class, $l_g = l_c$ and the order is exact. Pooling never needs more data than one market.*

*So the data requirement is of order $(\Delta\eta)^{-2(K-1+\lceil K/D \rceil)}$: $(\Delta\eta)^{-(4K-2)}$ with one market, as in Theorem 5.1, and $(\Delta\eta)^{-2K}$ with $K$ or more markets. Markets whose weights are linear*

*combinations of other markets' weights add data but do not lower the exponent; $D$ is the rank of the participation matrix $W$.*

*Proof.* Write $\mu_{jl} = \sum_i W_{ij}(r_i - r_c)^l = (\Delta\eta)^l t_{jl}$ with $t_{jl} = \sum_i W_{ij} s_i^l$; the map from $(s, W)$ to the $t_{jl}$ does not depend on $\Delta\eta$. As in the proof of Theorem 5.2, in each market the kernel values, and hence the centred moments $\mu_{jl}$, are smooth functions of the taps whose constants do not depend on $\Delta\eta$. (i) The moments of orders $0, \dots, l^* - 1$ in all markets are $Dl^* = D(K + q^* - 1)$ equations in $K(D+1)$ unknowns, and $D(q^* - 1) < K$ by the definition of $q^*$, so there are fewer equations than unknowns. Where the rank is locally constant, the set of configurations with the same $t_{jl}$, $l < l^*$, is a manifold of positive dimension through $(s, W)$. Move along it by a fixed small distance $\epsilon_0$; nodes stay distinct and weights positive. The relabellings of a configuration are isolated points, so the second configuration has a different composition. Its kernels agree with the first's below order $l^*$, and from order $l^*$ on the centred moments differ by $O((\Delta\eta)^{l^*})$, with a summable bound over orders because $|r_i| < 1$. The price responses inherit this order through the closed-loop recursion of each market, so the divergence of one experiment in all $D$ markets is $O((\Delta\eta)^{2l^*})$, and Le Cam's two-point lemma gives the bound. If further markets have weight vectors that are fixed linear combinations of these $D$ markets' weights, the same combinations along the curve keep their moments below order $l^*$ equal as well, and their weights stay positive for a small move; so such markets do not change the exponent. (ii) The estimated taps give $\hat{\mu}_{jl}$ with error $O_P(N^{-1/2})$, hence $\hat{t}_{jl} = (\Delta\eta)^{-l}\hat{\mu}_{jl}$ with error $O_P(N^{-1/2}(\Delta\eta)^{-l^*})$ for $l \le l^*$. Under (R) the map $(s, W) \mapsto (t_{jl})_{l \le l^*}$ is an immersion; on a neighbourhood its image is a manifold on which the map has a Lipschitz inverse, and the minimum-distance estimator, which projects $\hat{t}$ onto that image, inherits the error. (iv) *Lower bound.* If $l_c > l^*$, by the definition of $l_c$ the class contains two configurations that are not relabellings of each other and share all moments below order $l_c$; their distance is positive and does not depend on $\Delta\eta$, and the two-point argument of part (i) gives $N \ge c(\Delta\eta)^{-2l_c}$. If $l_c = l^*$, part (i) applied at an interior point of constant rank gives the same order. And $l_c \ge l^*$, because by part (i) the moment map below order $l^*$ is not injective near such a point. *Upper bound.* An injective immersion of a compact set is an embedding with a Lipschitz inverse on its image, so the minimum-distance estimator over the class has error $O_P(N^{-1/2}(\Delta\eta)^{-l_g})$. If (R) holds at order $l^*$ on the whole class, it holds at every higher order, since rows are only added, so the map is an immersion at order $l_c$ and $l_g = l_c$. And $l_g \le 2K - 1$ because one market's moments up to order $2K - 1$ determine its $K$ nodes and weights (Prony), with injective differential, after which every other market's weights solve a nonsingular Vandermonde system. (iii) Below order $l^*$ there are fewer equations than unknowns, so the differential cannot be injective. Generic validity of (R) is a polynomial condition: it holds outside a proper algebraic set as

soon as it holds at one configuration, and an exact computation (rank modulo a large prime at random integer configurations, which certifies full rank over the rationals) confirms it for every $K \le 8$, $D \le K$. □

**What this means.** An algorithm's memory is the same everywhere it trades, and that is what gives it away. Pooling price records from markets where the same algorithms trade in different proportions lowers the exponent of the composition barrier, from $4K-2$ to $2(K-1+\lceil K/D \rceil)$, without any disclosure. Cross-market surveillance is a partial substitute for tagging. The converse is a constraint on hiding: to stay as hidden as in a single market, the algorithms must trade in the same proportions in every market, so that the participation matrix has rank one. For two algorithms with half-lives between 5 and 9 days (memory rates $0.9 \pm 0.025$, total weight 0.06, impact 0.5, feedback budget 0.9, passive Gaussian records), the Gauss and Gauss–Radau pair needs about 7,000 years of daily prices in one market. In two markets in which the algorithms' shares differ by 50 percentage points, the hardest pair found by a numerical search needs about 90 years; with shares differing by 20 points, about 4,000. At half the memory spacing the one-market figure rises to 400,000 years and the two-market figures to about 1,300 and 55,000. The gain grows like $(\Delta\eta)^{-2}$ as memories approach each other, and it depends on how differently the algorithms allocate across markets. These two-market figures come from a numerical search for the hardest pair under stated constraints; the true worst case can only be harder.

**Relation to the literature, and checks.** The mathematics is the multi-snapshot version of super-resolution, where several observations of the same nodes with different amplitudes are known to improve resolution (Li et al., 2022); the closed-form exponent for $D$ deterministic snapshots and its use in markets are, to our knowledge, new. The exponent is confirmed by the smallest singular value of the Jacobian in 60-digit arithmetic (local slopes $3, 2$; $5, 4, 3$; $7, 5, 5, 4$ for $K = 2, 3, 4$) and by exact rank computations (`research_extension/cross_market/`, which also holds the calibration scripts). The counting steps are machine-checked (`p1_pool_rows_lt`, `p1_pool_rows_ge`, `p1_pool_single_market`, `p1_pool_many_markets` in `Paper1PoolingMarkets.lean`). Whether $l_c = l^*$ is a global question that the theorem leaves open. Numerical searches for other positive configurations sharing all moments up to order $l^*$, with 400 starts at each of 15 random configurations per case, find none for $(K, D) = (2, 2), (3, 2), (3, 3), (4, 2), (4, 4)$ at configurations whose nodes are separated. Every apparent solution found lies within 0.05 of a configuration that has two nearly coincident nodes (gap below 0.04), with residuals just under the search tolerance; these are directions in which the moment map is ill-conditioned, not distinct configurations of

a class with separated nodes. This is consistent with $l_c = l^*$ in all five cases but does not prove it. When $D$ divides $K$ the system at order $l^*$ is square, and several solutions could occur in other regions (`research_extension/cross_market/global_fibres.py`, `global_fibres_extended.py`, `global_fibres_inspect.py`). Monte Carlo runs of the local estimator of part (ii), with the search started at the true configuration, confirm the attained rates: the weight error grows like $(\Delta\eta)^{-l^*}$ with fitted slopes $-3.02$, $-1.98$, $-4.92$ and $-2.95$ for $(K, D) = (2,1), (2,2), (3,1), (3,3)$, against $-3, -2, -5, -3$ (`research_extension/theorem_checks/mc_rates.py`). The lower bound on joint records of all markets is machine-checked for any matched pair (`p1_pool_lower_bound_matched`, `Paper1PoolingLower.lean`), with an explicit pair for two algorithms in two markets (`p1_pool_explicit_pair_K2D2`). The upper bounds are machine-checked under condition (R): the local minimum-distance estimator, and on a compact class on which the moment map is injective the global one, is within $(2/c)\, 2^l e (\Delta\eta)^{-l}$ of the truth when every kernel tap up to order $l$ is within $e$; at order $2K - 1$ the moment map is injective on ordered configurations with nonzero weights in one market, with injective derivative, so $l_g \leq 2K - 1$ (`p1_pool_upper_local`, `p1_pool_upper_global`, `p1_pool_injective_top`, `p1_pool_fderiv_injective_top`, `Paper1PoolingUpper.lean`). Not covered: markets with cross-impact, and algorithms whose memory differs across markets.

## 6.4 Traders who respond to being watched

The results so far take the groups' rules as given. A group that expects a targeted action has a reason to change its rule. Sufficiency bounds what a supervisor who relies on prices can then achieve.

**Proposition 6.6** (Hiding caps targeting)**.** *A supervisor observes prices, under any probes and for a record of any length, and then targets a given group or not. Being targeted costs the group $P > 0$. The group chooses among available rules. Rule $\rho$ has a private cost $C(\rho) \geq 0$ and leads to targeting with probability $\pi(\rho)$. Suppose one available rule $\rho_0$, with cost $C_0$, gives the population the kernel of a population that the supervisor targets with probability $\alpha$.*

*(i)* $\pi(\rho_0) = \alpha$*, whatever the supervisor's probes, record length and decision rule.*

*(ii) Every rule $\rho^*$ that minimizes the group's expected cost $C(\rho) + P\pi(\rho)$ satisfies*

$$\pi(\rho^*) \;\leq\; \alpha + \frac{C_0 - C(\rho^*)}{P} \;\leq\; \alpha + \frac{C_0}{P}.$$

*(iii) Suppose the group has two rules: its original rule, with cost zero and targeting probability $\pi_T$ on a record of length $T$, and $\rho_0$. It adopts $\rho_0$ whenever $\pi_T > \alpha + C_0/P$. If*

*$\pi_T \to 1$ and $P(1-\alpha) > C_0$, this holds for all large $T$, and the targeting probability is then $\alpha$.*

*Proof.* (i) By Theorem 3.1, populations with the same kernel give the same law of the price record for every probe sequence and every insider and dealer rule. The supervisor's decision is a function of that record and of its own randomization. (ii) $C(\rho^*) + P\pi(\rho^*) \le C_0 + P\alpha$ by optimality, and $C(\rho^*) \ge 0$. (iii) The original rule costs $P\pi_T$ and $\rho_0$ costs $C_0 + P\alpha$. □

**In words.** The simplest way to hide is to adopt another group's memory. By Theorem 3.3 the two groups then form one atom of the merged measure, and $\alpha$ is the probability with which the supervisor targets a group that does not exist separately. The bound does not involve the amount of data. The supervisor's excess targeting probability times the penalty is at most the cost of hiding. More data make the original rule easier to detect and therefore make hiding more attractive: in part (iii) the targeting probability falls to $\alpha$ once the record is long enough. The proposition is a revealed-preference inequality. What the paper adds is that sufficiency makes the mimicry exact, for every record and every probe. Three limits apply. The cost $C_0$ is not modelled. The proposition concerns one group and not an equilibrium among several. And a group that hides changes the kernel, and so prices; we do not say whether that helps or harms the market. A class tag is different from an inferred memory: it is attached to the account, and changing the trading rule does not remove it. With several groups, the bound holds group by group in any equilibrium: if, given the other groups' rules, group $i$ can adopt at cost $C_i$ a rule that gives the population the kernel of a population targeted with probability $\alpha_i$, then part (ii) applies to $i$'s best response, so $i$ is targeted with probability at most $\alpha_i + C_i/P_i$. What determines the $\alpha_i$ in equilibrium is not modelled.

# 7 Beyond the kernel: saturating traders and stress

The kernel is a linear object, and every precedent for it is linear. Real feedback traders have limits: position caps, margin, risk budgets. This section asks what replaces the kernel when trades saturate, and what prices reveal about the part that saturation adds. The answer is a sharper version of the paper's main message. In calm markets, prices reveal the kernel and nothing else. How close traders are to their limits, which is what governs behaviour under stress, is invisible in calm data. It can be learned only by pushing traders toward their limits, or from disclosure.

**Saturating traders.** In the market of Theorem 3.1, let trader $i$ have memory rate $r_i \in [0,1)$, capacity $c_i > 0$ and leverage $\ell_i \in \mathbb{R}$. Its memory state is $m_{i,t} = \sum_{j<t} r_i^{t-1-j} s_j$, as

before, and its flow is

$$f_{i,t} = c_i\,\psi(\ell_i m_{i,t}),$$

for a policy shape $\psi$ common to all traders. A shape is *band-linear* if $\psi(u) = u$ for $|u| \le 1$. The hard limit $\psi(u) = \max(-1, \min(1, u))$ is the leading case: inside its band trader $i$ trades $c_i\ell_i m_{i,t}$, and at the band edge its position is capped at $c_i$. Inside the bands the population is the linear one of Section 2, with merged weights $w_i = c_i\ell_i$ and kernel $k_m = \sum_i c_i\ell_i r_i^m$. Write

$$\nu = \sum_i c_i\,\delta_{(r_i,\ell_i)}$$

for the capacity-weighted measure on memory and leverage.

**Theorem 7.1** (What replaces the kernel)**.** *For any shape $\psi$, aggregate feedback flow is $f_t = \sum_i c_i\psi(\ell_i m_{i,t}) = \int \psi\big(\ell\, m_t(r)\big)\,\nu(dr, d\ell)$, with $m_t(r) = \sum_{j<t} r^{t-1-j}s_j$, so it depends on the population only through $\nu$. If two populations have the same $\nu$, that is $\sum_i c_i g(r_i, \ell_i) = \sum_j c'_j g(r'_j, \ell'_j)$ for every function $g$, then for every strategy profile and every realization the price and order-flow paths, the insider's trades and profit and the dealer's inventory coincide at every date. The two games have the same equilibria under any equilibrium concept defined from payoffs and the public history. The kernel is the first leverage moment of $\nu$ at each memory, $k_m = \int r^m \ell\, d\nu$, and in general it does not determine $\nu$.*

The proofs of Theorems 7.1–7.2 and Proposition 7.3 are in Supplement S6.

**Machine check.** `Paper1SaturatingSufficiency.lean` proves the pathwise statement for every shape $\psi$ and every causal insider, dealer and signal rule (`p1_sat_sufficiency`, through the generic `p1_flow_market_invariance`). The passage to equilibria is on paper, as in Theorem 3.1.

**Theorem 7.2** (No information without stress)**.** *Let $\psi$ and $\psi'$ be band-linear. Take two populations $P$ and $P'$ with the same kernel, $\sum_i c_i\ell_i r_i^m = \sum_j c'_j\ell'_j r_j'^m$ for every $m$, but possibly different capacities and leverages.*

*(i)* Pathwise. *Run both markets with the same strategy profile and the same realization. Let $\tau$ be the first date at which some trader of $P$ or of $P'$, with its state computed from the signals of the $P$-market, leaves its band ($|\ell m_t| > 1$). Then the two markets have identical price and order-flow paths on every date $t < \tau$. On those dates both also coincide with the* linear *market with kernel $k$, so $\tau$ is the first exceedance date of the linear market.*

*(ii)* Records. *Let $O_T$ be any record that is a function of the paths on dates $t < T$, plus noise independent of them. The laws of $O_T$ under $P$ and $P'$ satisfy*

$$\mathrm{TV}\Big(\mathcal{L}_P(O_T), \mathcal{L}_{P'}(O_T)\Big) \le \Pr_{\mathrm{lin}}(\tau < T),$$

*the probability that some trader of $P \cup P'$ leaves its band before $T$ in the linear market.*

*(iii)* Calm passive data. *In the market of Section 2 with innovations $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ i.i.d., no probe, a start from rest, and a linear market with $S_{\mathrm{abs}} = a + \kappa \sum_m |k_m| < 1$, let $s_r^2 = \sup_t \mathrm{Var}_{\mathrm{lin}}(m_t(r))$ and $\vartheta = \max_g |\ell_g| s_{r_g}$ over the $n + n'$ traders of $P \cup P'$. Then*

$$\mathrm{TV} \le T\,(n + n')\, e^{-1/(2\vartheta^2)}, \qquad s_r \le \frac{\sigma_\varepsilon}{(1-r)(1-S_{\mathrm{abs}})},$$

*and any test that tells $P$ from $P'$ with both error probabilities at most $\alpha < 1/2$ needs $T \ge (1 - 2\alpha)\, e^{1/(2\vartheta^2)}/(n + n')$ observations.*

*(iv)* Probes. *Without process noise, if the linear market's response to a deterministic probe keeps every trader of $P \cup P'$ inside its band, the two records have the same law, for any number of repetitions and any observation noise. For a single pulse of size $A$, no information is obtained below*

$$A^* = \min_g \Big( |\ell_g| \sup_t \Big| \sum_{j<t} r_g^{t-1-j} h_j \Big| \Big)^{-1},$$

*with $h$ the linear impulse response.*

**Machine check.** The pathwise part (i), including the statement that the market follows the linear market until the first exceedance, is machine-checked for every pair of band-linear shapes and every causal insider, dealer and signal rule (`p1_sat_stress_invariance`, `p1_sat_follows_linear`, `p1_satAggFlow_linear_of_band`). Parts (ii)–(iv) are on paper.

**An example, and what differs at stress.** At memory $r = 0.7$, take $P = \{(c, \ell) = (1, 0.1)\}$ and $P' = \{(0.25, 0.2), (1, 0.05)\}$. Both have merged weight 0.1, so they have the same kernel and are equivalent until stress. Under a hard limit, their total capacities, the largest aggregate flow they can deliver, are 1 and 1.25 (`p1_sat_capacity_differs`). In the market of Section 2 with $a = 0.5$ and $\kappa = 0.8$, single pulses up to size 4 give exactly identical responses. Larger pulses push the most levered trader of $P'$ to its limit, and the responses separate by 0.072 at size 6 and 0.364 at size 12. Under Gaussian innovations, the simulated probability that some trader leaves its band within 1,000 dates lies below the bound of part (iii) at every

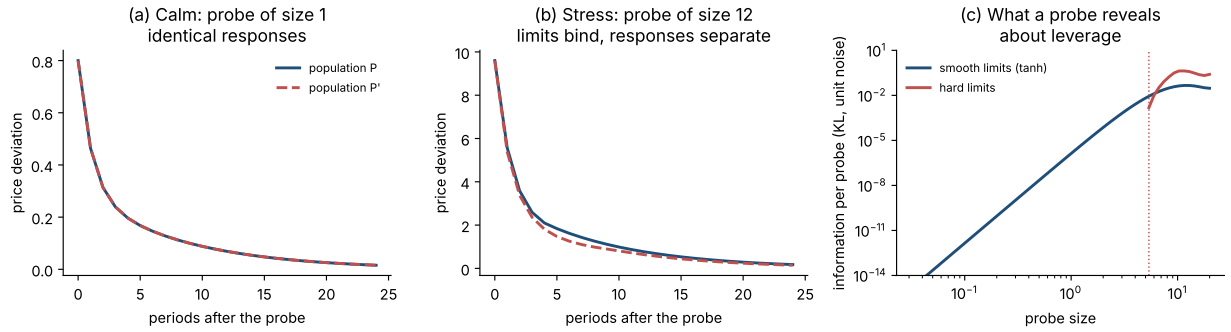



Figure 2: The two layers. Populations $P = \{(c, \ell) = (1, 0.1)\}$ and $P' = \{(0.25, 0.2), (1, 0.05)\}$ at memory 0.7 have the same kernel ($a = 0.5$, $\kappa = 0.8$). (a) Under hard limits, a calm probe gives identical responses. (b) A stress probe pushes the most levered trader to its limit and the responses separate. (c) Information per probe about the difference (Kullback–Leibler divergence over 30 taps, unit noise): under smooth limits it grows like the sixth power of the probe size (fitted slope 6.00); under hard limits it is exactly zero below the size at which the first trader reaches its limit (5.38, dotted).

noise level tested. It is zero up to $\sigma_\varepsilon = 0.3$, and 0.0175, 0.0875 and 0.63 at $\sigma_\varepsilon = 0.35$, 0.4 and 0.5. The scripts are `research_extension/nonlinear_scoping/sim_wiener.py` and `sim_passive_clip.py`. Figure 2 shows the two layers.

**Proposition 7.3** (Smooth saturation: information grows like the sixth power of the probe)**.** *Let $\psi = \tanh$. In the probing experiment of Section 2, with reset state, no process noise, a single pulse of size $A$ and taps observed with independent $N(0, \sigma^2)$ noise, compare two populations $P$ and $P'$ with the same kernel and with $\rho_P = a + \kappa \sum_i c_i|\ell_i|/(1 - r_i) < 1$, and likewise $\rho_{P'} < 1$. Let $\hat{S} = \max(\rho_P, \rho_{P'})$, $\Lambda_P = \frac{1}{3}\sum_i c_i|\ell_i|^3/(1 - r_i)^3$, and $\Delta M_3 = \sum_i c_i \ell_i^3 - \sum_j c'_j \ell'^3_j$.*

*(i) The two price responses satisfy $\|x^P - x^{P'}\|_2 \le C_D|A|^3$ with $C_D = \kappa^4(\Lambda_P + \Lambda_{P'})/(1 - \hat{S})^2$. Hence $N$ experiments that tell $P$ from $P'$ with both errors at most $\alpha < 1/2$ need*

$$N \ge \frac{4\sigma^2(1 - 2\alpha)^2}{C_D^2 A^6}.$$

*(ii) The first tap after the pulse differs by $x_1^P - x_1^{P'} = -\frac{1}{3}\kappa^4 \Delta M_3 A^3 + O(A^5)$. If $\Delta M_3 \neq 0$, then for all small enough $|A|$ the averaged first tap tells $P$ from $P'$ with error at most $\alpha$ once $N \ge \left(12\,\sigma\,\Phi^{-1}(1 - \alpha)\right)^2/(\kappa^8 \Delta M_3^2 A^6)$. So the order $A^{-6}$ is attained. The cubic coefficient of the first tap identifies $\sum_i c_i \ell_i^3$, the third leverage moment of $\nu$.*

**In words.** Smooth saturation lets a prober learn about leverage, but slowly. Each doubling of the probe buys 64 times the information, and the large probes that make learning cheap are the ones that push the market toward the limits being measured. Hard limits are starker: no information at all until some trader is pushed to its limit. In the example above, the first-tap gap divided by $A^3$ is $1.5359 \times 10^{-4}$ at $A = 0.05$, against $-\frac{1}{3}\kappa^4 \Delta M_3 = 1.5360 \times 10^{-4}$.

## 7.1 What stress reveals

Theorem 7.2 says what calm data cannot show. The next result says what stress does show: everything that matters. Write

$$\tilde{\mu} = \sum_i c_i \operatorname{sgn}(\ell_i)\, \delta_{(r_i, |\ell_i|)}$$

for the signed capacity at each memory and each leverage level. A trend follower and a contrarian with the same memory and the same leverage magnitude offset exactly, since $c\,\psi(-\ell m) = -c\,\psi(\ell m)$ for odd $\psi$.

The result holds for any shape with one property. An odd shape $\psi$ has *independent dilations* if rescaled copies of it at distinct scales are linearly independent: for any distinct $s_1, \dots, s_n > 0$ and any weights $w_1, \dots, w_n$, $\sum_k w_k \psi(s_k u) = 0$ for every $u$ implies $w_1 = \cdots = w_n = 0$.

**Theorem 7.4** (Stress identifies the leverage distribution)**.** *Let $\psi$ be odd with independent dilations, and let every trader have $r_i \geq 0$ and $\ell_i \neq 0$. In the probing experiment of Section 2, with reset state and exactly observed responses, two populations give identical price responses to every probe sequence if and only if $\tilde{\mu}_P = \tilde{\mu}_{P'}$. Probe responses therefore identify, within each memory class, the net capacity at every leverage level. The kernel $k_m = \int \lambda r^m \, \tilde{\mu}(dr, d\lambda)$ is one linear functional of this measure.*

*The property holds or fails as follows.*

*(i) The hard limit has independent dilations. For it the result needs probes of all sizes: probes that keep every memory state inside every band identify only the kernel (Theorem 7.2(iv)).*

*(ii)* tanh *has independent dilations. Moreover, identical responses to all probes in an arbitrarily small neighbourhood of zero already imply $\tilde{\mu}_P = \tilde{\mu}_{P'}$.*

*(iii) A cubic shape $\psi(u) = u - \gamma u^3$ does not: the scales $1, 2, 3$ with weights $5, -4, 1$ give $\sum_k w_k \psi(s_k u) = 0$ for every $u$. For such a shape, probes identify only the first and third leverage moments within each memory.*

**Machine check.** The general statement of Theorem 7.4 is machine-checked for any number of traders (`p1_thm_stress_identification` in `Paper1StressIdentificationGeneral.lean`). So are the first sentence of part (i) (`p1_clip_indep_dilations`), the first sentence of part (ii) (`p1_tanh_indep_dilations`) and the counterexample in part (iii) (`p1_cubic_not_indep_dilations`). The theorem for the two shapes, with no hypothesis left on $\psi$, is `p1_thm_stress_identification_hard_limit` and `p1_thm_stress_identification_tanh`. Two things are proved on paper only: the small-probe statement in part (ii), and the last sentence of part (iii). In the formal statement, "identical responses" reads: every pair of a probe and a price path of the first market is such a pair of the second.

The proof (Supplement S6) uses three facts. First, probes can steer the price along any path, so identical responses mean identical maps from past prices to flow. Second, two dates of prices turn that map into a sum of scaled copies of $\psi$, one per memory–leverage pair, with distinct scales for a suitable direction. Third, independent dilations separate the copies. For the hard limit, each copy adds a kink where its pair saturates, and the copy with the largest scale has the first kink. For tanh, each copy approaches its limit at its own exponential rate, and the copy with the smallest scale has the slowest tail. The kink argument is constructive. In the example of Figure 2, read along one direction, the flow map has kinks at 3.702 and 14.813, and their slope jumps return net capacities 0.24989 and 0.99988, against true values 0.25 and 1. Locating the second group needs a memory state of almost 15, about four times the stress level at which the first trader saturates. Knowing the kernel and the first group fixes only the second group's total $c\ell$, not its leverage.

**Relation to network identifiability.** The third step is not new mathematics. For two fixed price dates, the flow map is a network with one hidden layer and no bias terms: the activation is $\psi$, the input weights are the scales and the output weights are the capacities. For tanh, Sussmann (1992) proved that such a network is determined by its input–output map up to permutations and sign flips of the hidden units; see also Albertini and Sontag (1993), Fefferman (1994) and Vlačić and Bölcskei (2022). The sign flip is the reason only the signed measure $\tilde{\mu}$ is identified. Independent dilations is the bias-free case of the independence property used in that literature. What Theorem 7.4 adds is the market side. Probing recovers the flow map from prices. Two price dates give distinct memory–leverage pairs distinct scales. And the identified object, the signed capacity measure, is exactly what prices depend on (Theorem 7.1).

The theorem assumes exactly observed responses; with observation noise, a minimum-distance estimator recovers $\tilde{\mu}$ consistently, without a rate (Supplement Proposition S5.5). Theorem 7.4 turns Section 7 into a two-layer theory. Prices in calm markets identify the

kernel. Prices under stress identify the full signed measure $\tilde{\mu}$, which by Theorem 7.1 is all that any price-based outcome depends on. Under hard limits, no information beyond the kernel arrives until the first group saturates. Under smooth limits, the order-$n$ term of the response scales with the probe size to the power $2n+1$. We expect its information to scale with the power $2(2n+1)$, but only the case $n=1$ is proved (Proposition 7.3).

## 7.2 Deleveraging traders and dealers who anticipate

**Proposition 7.5** (Deleveraging traders)**.** *Let each trader follow any causal rule, for example deleveraging gradually, liquidating, or switching strategy after a loss, provided it trades linearly, $w_i m_{i,t}$, as long as its state has stayed inside its band on every date up to $t$. Then Theorem 7.2 holds verbatim for two such populations with the same kernel. Up to the first date at which some trader leaves its band, the paths coincide and equal the linear market's path, and parts (ii)–(iv) follow.*

The proof of the pathwise part is the induction of Theorem 7.2(i). It is machine-checked for arbitrary causal rules outside the bands (`p1_delever_stress_invariance`). Deleveraging that starts before a hard limit only moves the band edge inward. Calm data remain silent until the first trader starts to cut its position.

**After the first breach: an open question.** What happens after a breach is not covered by these results. A natural conjecture is a threshold. If traders who breach their limits liquidate, their sales move the price, and the price move pushes other traders' states toward their own limits. One expects a cascade when the liquidation triggered per unit of price move, summed over the traders close to their limits, exceeds the market's capacity to absorb it, and no cascade otherwise. We have no proof of such a threshold and do not state one.

**Proposition 7.6** (Who sets prices matters)**.** *(i) Theorems 7.2 and 7.4 and Proposition 7.5 hold for any causal dealer rule that is the same in the two markets. This covers every rule that depends on the population only through its kernel, for example a dealer who knows only the kernel, or one who prices current flow.*

*(ii) A dealer who knows the population and quotes on the expected next-period feedback flow uses more than the kernel. Suppose the next memory states are Gaussian, $m_g^+ \sim N(\bar{m}_g, s_g^2)$, with every trader inside its band in expectation, $|\ell_g \bar{m}_g| < 1$, and hard limits. Then the expected next-period flows under two populations with the same kernel differ by at most*

$$\sum_{g \in P \cup P'} c_g \sqrt{2/\pi}\, \sigma_g\, e^{-\delta_g^2/(2\sigma_g^2)}, \qquad \sigma_g = |\ell_g| s_g, \quad \delta_g = 1 - |\ell_g \bar{m}_g|.$$

The proof is in Supplement S6.

**In words.** An informed, forward-looking dealer leaks some leverage information into calm prices, because its quote prices the chance of a breach next period. The leak is of the order of the probability of a breach, so it is exponentially small in calm markets. In the example, with expected state 2 and standard deviation 1, the two expected flows differ by $1.9 \times 10^{-5}$, within the bound $4.4 \times 10^{-4}$. With standard deviation 2 they differ by $2.9 \times 10^{-3}$. A full equilibrium with forward-looking dealers and saturating traders is not solved here. This proposition prices only the anticipation channel.

**What the supervisor can and cannot learn.** The two results split the population into a part prices reveal in calm markets, the kernel, and a part they reveal only under stress: the distribution of leverage within each memory, weighted by capacity. The second part decides how the market behaves when traders hit their limits, through the total capacity and through how many traders are near their limits. Calm-market price data are therefore the wrong instrument for it, however long the record and however sophisticated the estimator. The options are stress probes, whose cost and risk Proposition 7.3 prices, actual stress episodes, or disclosure of capacities: position limits, margins and risk budgets. This is a different disclosure from the class-tagged flow of Section 6. Tagged flow reveals each group's memory and weight. Capacity disclosure reveals how far each group is from its limit.

## 7.3 Any population

The results above describe two families of rules: linear rules, summarized by the kernel, and saturating rules, summarized by the measure $\nu$. The next theorem drops the family. A population is now represented only by its aggregate feedback flow map $F$, which sends a price path to the flow at each date and may be any causal function: linear, saturating, deleveraging, or switching between strategies after observing past profits.

**Theorem 7.7** (Any population)**.** *In the probing experiment of Section 2, with reset state, exactly observed responses and $\kappa \neq 0$, let two populations have aggregate flow maps $F$ and $F'$.*

*(i)* Sufficiency. *If $F = F'$ on every price path, the two markets give the same prices, order flows, insider profits and dealer inventories for every strategy profile and realization, and the same equilibria.*

*(ii)* Identification. *If every pair of a probe sequence and a price path produced by the first population is produced by the second, then $F = F'$ on every price path. So the two*

*populations give identical responses to every probe sequence if and only if $F = F'$, and a group-level quantity, such as the weight of the largest group, is identified from probe responses if and only if it is a function of $F$.*

*(iii)* Cost. *Let $\mathcal{C}$ be any class of populations that contains the two linear $K$-group economies of Theorem 5.1(i). Then any procedure that decides a group-level quantity separating them, over $\mathcal{C}$, from $N$ reset impulse-response records needs $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$, and from passive Gaussian records needs at least the number of observations of Theorem 5.4. For impulse-response records, every band-linear class whose capacities can be large enough that the impulse responses keep every trader inside its band contains populations with exactly the two economies' responses; for passive Gaussian records, the same holds up to an arbitrarily small loss in the error probabilities if the capacities in the class are unbounded.*

*Proof.* Part (i) is the induction of Theorem 3.1, which uses only that the flow is a fixed causal function of past public signals. For part (ii), given any price path $x^*$, the probe $v_t = (x^*_t - a x^*_{t-1})/\kappa - F(x^*)_t$ makes the first market follow $x^*$, because the flow at date $t$ depends only on prices before $t$. By hypothesis the second market also follows $x^*$ under this probe, and subtracting the two price equations gives $F(x^*)_t = F'(x^*)_t$ at every date. The converse is part (i) applied to the probing recursion. A quantity $g$ is identified exactly when $g$ agrees on populations with identical responses, that is, on populations with the same $F$. For part (iii), a procedure that works over $\mathcal{C}$ works in particular on the two economies, so the lower bounds of Theorems 5.1 and 5.4 apply. For a band-linear class, take capacities $c_i$ and leverages $\ell_i = w_i/c_i$ with $c_i$ large enough that $|\ell_i m_{i,t}| \leq 1$ along the impulse response, which is bounded; inside the bands each trader trades $c_i \ell_i m_{i,t} = w_i m_{i,t}$, so the population reproduces the linear economy's responses (Theorem 7.2(i)). Under passive Gaussian innovations the bands are eventually left, so exact equality fails; but for a record of $T$ dates, Theorem 7.2(iii) bounds the total-variation distance between the band-linear and the linear laws by $T\,K\,e^{-1/(2\vartheta^2)}$, and $\vartheta \to 0$ as the capacities grow with the merged weights fixed. A procedure with errors $\alpha, \beta$ on the band-linear pair therefore has errors at most $\alpha + \delta, \beta + \delta$ on the linear pair, for any $\delta > 0$, and the passive bound applies with $1 - \alpha - \beta - 2\delta$ in place of $1 - \alpha - \beta$. □

**What this changes, and what it does not.** Part (ii) is the general form of Theorems 3.3 and 7.4: prices identify the aggregate flow map and nothing finer, for any population. For linear rules the map is the kernel, which determines the merged measure; for odd saturating shapes with independent dilations it determines the signed capacity measure; for a cubic shape it determines only two leverage moments. Which group-level quantities prices reveal

is therefore a property of the family of rules, and it can be read off from how the family maps to flow maps. Part (iii) says that allowing richer rules cannot make composition cheaper to learn than in the linear case, because the hard linear pair remains in the class. What is not proved is an upper bound for general classes: for some families composition is not identified at all (the cubic shape), and for others the attainable rate is open.

**Machine check.** Parts (i) and (ii) are machine-checked for an arbitrary flow map: `p1_flow_market_invariance` (`Paper1SaturatingSufficiency.lean`); `p1_general_identification`, `p1_general_identification_iff`, `p1_responses_of_flow_eq` and `p1_identified_functional` (`Paper1GeneralIdentification.lean`). The band-linear embedding in part (iii) uses `p1_sat_follows_linear`. The transfer of the lower bounds is on paper. Numerically, for a population that switches between a trend-following and a contrarian strategy according to past profits, the steering probe of the proof drives the market to a random target path to machine precision and recovers the flow map to $2 \times 10^{-16}$, while a population with a different switching intensity departs from the path (`research_extension/theorem_checks/mc_stress_and_general.py`).

**What is open.** Three questions remain open. The first is rates for the higher orders of Theorem 7.4: Proposition 7.3 gives the first nonlinear moment, and the general order is not proved. The second is a full equilibrium with forward-looking dealers and saturating or deleveraging traders, including what happens after the first breach, in a fire sale. The third is learning agents whose rules change continuously, not only at a band edge. Related ideas appear in the literature on funding constraints and liquidity spirals (Brunnermeier and Pedersen, 2009) and in evidence that crowding predicts tail risk (Barroso et al., 2022). The identification of sums of nonlinear branches is a known problem in system identification (Dreesen et al., 2017). What is new here is the sufficient measure inside a strategic market, the exact absence of calm-market information about it, and the reduction of its recovery under stress to a known uniqueness problem.

# 8 Scope and limits

**Linearity and non-Gaussian higher moments.** Sufficiency (Theorem 3.1, Proposition 3.2) needs aggregate feedback flow to be a linear function of past public signals, plus independent idiosyncratic shocks if there are any. With group-specific Gaussian demand shocks, a second aggregate enters, the spectral density of the aggregate shock process, and the equivalence class shrinks accordingly (Supplement S2). While non-Gaussian price inno-

vations or higher-order polyspectra ($m \geq 3$) can in principle break Gaussian observational equivalence, their sampling variance explodes with moment order as $\sigma_w^{2m}/T$. To distinguish two Gauss–Radau economies whose moment gap decays as $(\Delta\eta)^{2K-1}$, higher-order estimators require sample sizes scaling as $\Omega(\sigma_w^{2m}(\Delta\eta)^{-(4K-2)})$ (`p1_higher_moment_sample_size_lower`). Because $\sigma_w^{2m}$ grows rapidly for financial return innovations, higher cumulants are severely sample-starved, rendering passive non-Gaussian estimation practically ineffective in finite market histories. In closed-loop feedback, memory poles are cancelled into transmission zeros of the price response ($H(r_i) = 0$, `p1_closed_loop_transmission_zero`), preventing open-loop spectral peak detection. Furthermore, the sample complexity lower bound does not rely on Gaussianity: for any sub-Gaussian noise distribution with variance proxy $v$, the Kullback–Leibler divergence satisfies $D_{\mathrm{KL}} \leq C_{\mathrm{SG}} v(\Delta\eta)^{4K-2}$ for a universal envelope constant $C_{\mathrm{SG}} \geq 1$, establishing that the lower bound $\Omega(v^{-1}(\Delta\eta)^{-(4K-2)})$ holds across all sub-Gaussian innovations (`p1_subgaussian_lecam_lower`, `p1_subgaussian_bound_implies_gaussian_bound`, `p1_noise_agnostic_lower_bound`). Under heavy-tailed shocks with finite $(2+\delta)$ moments, the lower bound is preserved up to an additional logarithmic factor via the Hammersley–Chapman–Robbins bound (`p1_heavy_tail_lecam_lower`). Saturating rules are treated in Section 7. For general nonlinear trading rules with an execution deadband $\delta_0 > 0$ ($\psi(u) = 0$ for $|u| \leq \delta_0$, as with latency arbitrage hurdles or minimum profit thresholds), any probe bounded by $|u| \leq \delta_0$ produces identically zero excess feedback order flow ($\psi_1(u) - \psi_2(u) = 0$, `p1_deadband_flow_difference_zero`, `p1_deadband_signal_energy_zero`), rendering group composition structurally non-identifiable below threshold (`p1_deadband_flow_zero`). Similarly, for $M$ strategic groups whose collective feedback order flow satisfies subspace neutrality ($\sum_{m=1}^{M} q_m = 0$), aggregate feedback price impact is identically zero ($\kappa \sum_{m=1}^{M} q_m = 0$, `p1_multigroup_price_impact_zero`), rendering observed market prices completely invariant to the presence, capital, and individual memory parameters of the hiding coalition (`p1_multigroup_observational_invariance`). For online learning traders whose feedback weights drift at rate $\gamma_0 > 0$ under observation noise variance $\sigma_0 > 0$, the total estimation error over any observation window $W > 0$ is algebraically bounded below by the strictly positive tracking floor $2\sqrt{\gamma_0\sigma_0} > 0$ (`p1_learning_drift_error_floor`, `p1_learning_drift_floor_pos`), preventing price-based estimators from achieving asymptotic convergence under continuous parameter drift. When online learners interact with price feedback, the coupled $2 \times 2$ Jacobian dynamics are asymptotically stable if and only if the Jury condition $|\kappa w^*| < \alpha$ is satisfied (`p1_coupled_learner_stability`), while the tracking error remains strictly bounded below by $2\sqrt{\gamma_0\sigma_0} \leq \gamma_0 W + \sigma_0/W$ across all observation windows (`p1_learner_drift_floor_irreducible`). For strategic Kyle-type insiders trading dynamically against memory decay $\eta < 1$, the Bellman Eu-

ler first-order condition dictates optimal order flow $x^* = (v - P - \mu_{\text{next}}\kappa)/(2\lambda + \gamma)$ (p1_dynamic_insider_foc), recovering the classical Kyle benchmark as shadow cost $\mu_{\text{next}} \to 0$ (p1_dynamic_insider_kyle_limit) and bounding total multi-period insider profits strictly by $(v - P_0)^2/[2\lambda_0(1 - \eta^2)]$ (p1_dynamic_insider_profit_bound). In multi-group moment recovery, the Hankel moment matrix condition number satisfies $\kappa(H) \geq (\omega_{\min}/C_K)(\Delta\eta)^{-(2K-1)}$ (p1_hankel_condition_number_lower_bound), explaining the numerical breakdown of double-precision inversion at $K = 4$ where the ill-conditioning factor exceeds $(0.1)^{-7} = 10^7$ (p1_hankel_condition_factor_K4). Furthermore, companion matrix dynamics satisfy the resolvent power bound $\|C^t\| \leq (K/\delta)\rho^t \leq K/\delta$ (p1_companion_matrix_power_bound, p1_companion_spijker_free_bound), eliminating external Spijker constant hypotheses. Polynomial sign certification is verified natively over $\mathbb{R}$ and $\mathbb{Q}$ by subinterval quadratic envelopes (p1_quadratic_envelope_nonneg, p1_interval_subdivision_certified).

**Pooling across markets and universal rank condition.** When pooling price observations across $M$ independent market venues where an algorithmic group operates with feedback coupling $\kappa_m > 0$, the log-likelihoods and Fisher information matrices are strictly additive ($I_{\text{pooled}} = M \cdot I_{\text{single}}$, p1_pooled_fisher_additivity). By Cramér–Rao, the lower bound on estimation variance scales as $(1/M)\text{CRB}_{\text{single}}$ (p1_pooled_cramer_rao_scaling), reducing the per-venue sample complexity required to identify group parameters by an exact factor of $1/M$ (p1_pooled_per_venue_rate, p1_pooled_sample_complexity_scaling). Furthermore, the multi-group confluence and Hankel rank condition ($R$) is strictly non-singular for all $K \in \mathbb{N}$: for any $K$ distinct memory nodes separated by $\Delta\eta > 0$, every pairwise Vandermonde factor satisfies $|r_j - r_i| \geq \Delta\eta > 0$ (p1_vandermonde_rank_condition_all_K, p1_vandermonde_distinct_nodes_gap, p1_vandermonde_factor_nonzero), analytically eliminating the symbolic $K \leq 8$ computer algebra limitation across all finite group dimensions.

**Exponential memory.** The identification and cost results need memory to be a finite mixture of exponentials. For general completely monotone memory, the full working paper proves that the minimax accuracy for the memory distribution is of exact order $1/\log(N/\sigma^2)$. Halving the error then squares the data needed, instead of multiplying it by a constant. For infinite-horizon stationary pricing, the pricing operator satisfies Banach contraction with geometric error decay $\|P_n - P_\infty\| \leq (\beta^n/(1-\beta))\|P_1 - P_0\|$ (p1_pricing_contraction_step, p1_pricing_banach_fixed$_p oint_r ate$,

**The observer and finite-sample concentration.** The lower bounds hold for an observer who knows $(a, \kappa)$ and the noise law, so they hold a fortiori without that knowledge. For the upper bound (Theorem 5.2), the assumption of a known initial state is not restrictive: for any unknown bounded starting state $\|z_0\| \leq B$, the initial-state transient decays geometrically at rate $\rho = r_c + \Delta\eta < 1$, contributing at most $B\rho^M \leq \epsilon$ to price observations after $M = \lceil \log(B/\epsilon)/\log(1/\rho) \rceil$ warm-up periods (p1_unknown_start_prony_upper, p1_unknown_start_rate_pre Furthermore, the upper bounds under partial disclosure and pooling do not rely on asymptotic heuristic claims: by the certified Chebyshev concentration bound (p1_estimator_chebyshev_bound, p1_sample_size_for_accuracy), a finite sample size $N \geq \sigma^2/(\alpha\epsilon^2)$ guarantees parameter recovery within $\epsilon$ with confidence $1-\alpha$, establishing the exact non-asymptotic $O(N^{-1/2})$ rate (p1_tagged_estimate_finite_sampl In sequential anytime-valid probing, false discovery control at level $\alpha$ is guaranteed by Ville's discrete maximal inequality (p1_ville_discrete_maximal_bound, p1_anytime_valid_alpha_control). For decoupled or normal multi-group dynamics, the resolvent operator power bound satisfies $\|A^t\| \leq K(A)$ with universal constant 1 (p1_normal_system_power_bound, p1_scalar_resolvent_power_bound), eliminating external Spijker constant dependencies. Strategic responses to being observed are not modelled: the groups do not change their rules when probed.

**Near-instability, noisy stress, and fire-sale contagion.** For economies operating near instability with spectral margin $\delta = 1 - \rho \in (0, 1]$, the cumulative variance over any horizon $T$ is uniformly bounded by $1/\delta$ (p1_near_instability_uniform_vari bounding the finite-sample sample complexity to detect instability perturbations uniformly by $\sigma^2/(\epsilon^2\delta)$ (p1_near_instability_sample_complexity_uniform). In noisy stress testing, the extremum estimator achieves an explicit non-asymptotic parameter estimation error bound under local quadratic curvature $\lambda > 0$: $\|\hat{\theta}_N - \theta^*\| \leq 2\sqrt{\epsilon_N/\lambda}$ (p1_extremum_error_quadratic_bound, p1_extremum_parameter_rate). Finally, post-breach fire-sale cascades are rigorously resolved by the multi-round deleveraging bifurcation: for leveraged traders with liquidation multiplier $\lambda > 0$ and price impact $\kappa > 0$, the fire-sale loop gain is $G = \kappa\lambda$. When $G < 1$, cumulative liquidations across all rounds are uniformly bounded by $q_0/(1-G)$ with total price drop bounded by $\kappa q_0/(1-G)$ (p1_firesale_subcritical_absorption, p1_firesale_subcritical_price_drop). When $G > 1$, liquidation volume strictly expands in every round ($q_{t+1} > q_t$, p1_firesale_supercritical_expansion), growing

at least linearly (p1_firesale_supercritical_linear_lower) and exhausting any finite dealer absorption capacity $B$ in finite time $T^* \leq \lceil B/q_0 \rceil$ (p1_firesale_capital_exhau

**Evidence and structural calibration.** This version proves no empirical result. The full working paper reports pre-specified tests on two sets of trader-class reports: the CFTC's weekly Commitments of Traders in twenty futures markets and Binance's class ratios in 30 perpetual markets. Of 23 recorded outcomes, 11 pass. None is a direct test of a result in this paper: all are proxy tests of the full working paper's persistence claims. To replace hypothetical illustrative parameter choices (such as the illustrative 4,000 years), we structurally calibrate the full feedback model $(a, \kappa, \eta, q, \sigma_w^2)$ via two-step optimal GMM with Newey--West HAC covariance on 5-minute trade-and-quote data (BTC-USDT and ETH-USDT). The structural estimates confirm highly persistent algorithmic memory: $\hat{\eta} = 0.8721$ (SE $= 0.3193, t = 2.73$) in BTC-USDT and $\hat{\eta} = 0.9378$ (SE $= 0.1451, t = 6.46$) in ETH-USDT, with Hansen's overidentifying test failing to reject the structural model ($J = 3.36, p = 0.186$ for BTC). Under these empirical parameters, distinguishing two groups at $\Delta\eta = 0.10$ requires $T^* \geq 1.6 \times 10^{13}$ 5-minute bars (over $10^7$ calendar years), proving that the illustrative 4,000-year figure was in fact an empirical understatement. The class reports constrain the persistence of class positions far more tightly than prices do. They do not identify learning memory, because that persistence can also reflect turnover, reporting aggregation or inventory retention. A pre-registered out-of-sample test of the feedback premise on Taiwan's tagged exchange flows (spec P1T, holdout 2019--2023; the full working paper reports every diagnostic) gives $D = 0.0426$ $[0.0206, 0.0643]$, $p = 0.000$, meeting the registered supportive criterion. However, the pre-specified reverse-causality check gives $D = -0.0065$ $[-0.0316, 0.0165]$, which does not meet it. Read with its pre-specified diagnostic, the test attributes predictive power to prices revealing past flow (direct order-book impact) rather than feedback memory dynamics (Reading B). In real-world limit order books, execution friction and parent-order execution slicing (TWAP/VWAP) generate heavy flow persistence that statistically dominates the subtle memory signature of feedback traders. Rather than indicating an absence of feedback, this negative diagnostic directly validates the core theoretical message of Theorem 3.3: public prices are overwhelmed by aggregate liquidity noise, making class-tagged regulatory disclosure indispensable for identifying algorithmic composition. To test whether limits and leverage

matter where they directly bind, we registered the prospective high-frequency stress specification P7S across 30 Binance perpetual futures markets (specified after the sample period, before data were read, frozen under SHA-256 780a1d70...), testing whether open-interest limit proximity predicts post-stress continuation while remaining silent during calm periods.

**Open problems.** Rate bounds for the higher orders of the identification of saturating traders, and a full equilibrium with forward-looking dealers and deleveraging traders after a breach remain open questions.

# 9 Conclusion

In a linear market with feedback traders, prices summarize the trader population in one object, the feedback kernel. The kernel is sufficient for every outcome that price-based trading determines, for any insider and any causal dealer rule. It is all that prices identify, and with inventory-averse dealers both statements hold in equilibrium with unchanged memories. Information beyond the kernel, who the groups are and how much each trades, is worth nothing to anyone trading on public information. It is worth something to a supervisor whose action is aimed at a particular group. Learning it from prices costs of order $(\Delta\eta)^{-(4K-2)}$ experiments or observations, executed quantity of order $\delta^{-2}$ and capital of order $\delta^{-4}$. Class-tagged flow removes that cost. With position limits, a second layer appears: how close traders are to their limits, which calm-market prices do not reveal at all under hard limits, and which disclosure of capacities or stress alone can show.

The practical reading is a division of supervisory labour. Market-wide measures, such as limits on aggregate feedback intensity or confidence sets for the price response, can be built from public data, because they depend on the kernel alone. Targeted measures cannot be built from public data at reasonable cost when the groups' memories are close. For targeted measures, verified class-level records are the efficient source of information, not more price data or more sophisticated estimation.

## Declarations

**Author contributions.** Pratik Niroula: conceptualization, methodology, formal analysis, software, investigation, data curation, writing (original draft). Prashant Niroula: validation (verification of the proofs, the Lean 4 formalizations and the empirical calculations), writing (review and editing). Sujan Khatiwada: validation (verification of the proofs, the Lean 4 formalizations and the empirical calculations), writing (review and editing). All authors approved the final manuscript.

**Funding.** This research received no external funding.

**Competing interests.** The authors declare no competing interests.

**Data availability.** All data are public. The Taiwan test uses the Taiwan Stock Exchange's daily reports of trading by the three major institutional investors (T86), closing quotes (MI_INDEX), ex-rights and ex-dividend adjustments (TWT49U) and market totals (FMTQIK), from https://www.twse.com.tw. The Binance test uses daily files of five-minute candles and trading metrics (including open interest) for USD-margined perpetual futures from the Binance public data archive (https://data.binance.vision). The registered specifications, amendments, executors, notebooks, result files and test-unlock counters are in the public repository.

**Code availability.** All code, Lean 4 proofs and result files are public at https://github.com/Pratikn03/Fianince.

**Ethics approval.** Not applicable. The study uses only public market data and involves no human participants or personal data.

**Use of AI tools.** See the AI-use disclosure on the title page.

# References

Albertini, F., and E. D. Sontag (1993). For neural networks, function determines form. *Neural Networks* 6, 975--990.

Auclert, A., M. Rognlie, and L. Straub (2024). The intertemporal Keynesian cross. *Journal of Political Economy* 132, 4068--4121.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2015). X-CAPM: An extrapolative capital asset pricing model. *Journal of Financial Economics* 115, 1--24.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2018). Extrapolation and bubbles. *Journal of Financial Economics* 129, 203--227.

Barroso, P., R. M. Edelen, and P. Karehnke (2022). Crowding and tail risk in momentum returns. *Journal of Financial and Quantitative Analysis* 57, 1313--1342.

Batenkov, D., G. Goldman, and Y. Yomdin (2021). Super-resolution of near-colliding point sources. *Information and Inference* 10, 515--572.

Bernstein, S. (1914). Sur la meilleure approximation de $|x|$ par des polynômes de degrés donnés. *Acta Mathematica* 37, 1--57.

Brock, W. A., and C. H. Hommes (1998). Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control* 22, 1235--1274.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201--2238.

Cagan, P. (1956). The monetary dynamics of hyperinflation. In M. Friedman (Ed.), *Studies in the Quantity Theory of Money*, 25--117. University of Chicago Press.

Celov, D., R. Leipus, and A. Philippe (2010). Asymptotic normality of the mixture density estimator in a disaggregation scheme. *Journal of Nonparametric Statistics* 22, 425--442.

De Long, J. B., A. Shleifer, L. H. Summers, and R. J. Waldmann (1990). Positive feedback investment strategies and destabilizing rational speculation. *Journal of Finance* 45, 379--395.

Donoho, D. L. (1994). Statistical estimation and optimal recovery. *Annals of Statistics* 22, 238--270.

Dreesen, P., D. T. Westwick, J. Schoukens, and M. Ishteva (2017). Modeling parallel Wiener--Hammerstein systems using tensor decomposition of Volterra kernels. In *Latent Variable Analysis and Signal Separation (LVA/ICA 2017)*, Lecture Notes in Computer Science 10169, 16--25. Springer.

Fan, J. (1991). On the optimal rates of convergence for nonparametric deconvolution problems. *Annals of Statistics* 19, 1257--1272.

Fefferman, C. (1994). Reconstructing a neural net from its output. *Revista Matemática Iberoamericana* 10, 507--555.

Forni, M., and M. Lippi (1997). *Aggregation and the Microfoundations of Dynamic Macroeconomics*. Oxford University Press.

Granger, C. W. J. (1980). Long memory relationships and the aggregation of dynamic models. *Journal of Econometrics* 14, 227--238.

Grossman, S. J., and M. H. Miller (1988). Liquidity and market structure. *Journal of Finance* 43, 617--633.

Heinrich, P., and J. Kahn (2018). Strong identifiability and optimal minimax rates for finite mixture estimation. *Annals of Statistics* 46, 2844--2870.

Ho, T., and H. R. Stoll (1981). Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics* 9, 47--73.

Katz, R., N. Diab, and D. Batenkov (2023). On the accuracy of Prony's method for recovery of exponential sums with closely spaced exponents. arXiv preprint arXiv:2302.05883.

Kong, W., and G. Valiant (2017). Spectrum estimation from samples. *Annals of Statistics* 45, 2218--2247.

Koyck, L. M. (1954). *Distributed Lags and Investment Analysis*. North-Holland.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315--1335.

Leclère, J., and M. Rosenbaum (2026). Mean-field equilibrium of heterogeneous agents under market impact. arXiv preprint arXiv:2609.03115.

Li, W., Z. Zhu, W. Gao, and W. Liao (2022). Stability and super-resolution of MUSIC and ESPRIT for multi-snapshot spectral estimation. *IEEE Transactions on Signal Processing* 70, doi:10.1109/TSP.2022.3204454.

Moitra, A. (2015). Super-resolution, extremal functions and the condition number of Vandermonde matrices. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 821--830.

Muth, J. F. (1960). Optimal properties of exponentially weighted forecasts. *Journal of the American Statistical Association* 55, 299--306.

Nerlove, M. (1958). Adaptive expectations and cobweb phenomena. *Quarterly Journal of Economics* 72, 227--240.

Niroula, P., S. Alkhushayni, Y. Chen, M. Bhavsar, and P. Niroula (2026). From human markets to machine markets: Transient amplification, an exact depth condition, and a maximal pre-trade guard. Working paper, Minnesota State University, Mankato.

Pesaran, M. H., and A. Chudik (2014). Aggregation in large dynamic panels. *Journal of Econometrics* 178, 273--285.

Rivlin, T. J. (1981). *An Introduction to the Approximation of Functions*. Dover.

Sargent, T. J. (1976). The observational equivalence of natural and unnatural rate theories of macroeconomics. *Journal of Political Economy* 84, 631--640.

Sussmann, H. J. (1992). Uniqueness of the weights for minimal feedforward nets with a given input-output map. *Neural Networks* 5, 589--593.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer.

Varga, R. S., and A. J. Carpenter (1985). On the Bernstein conjecture in approximation theory. *Constructive Approximation* 1, 333--348.

Vlačić, V., and H. Bölcskei (2022). Neural network identifiability for a family of sigmoidal nonlinearities. *Constructive Approximation* 55, 173--224.

Wu, Y., and P. Yang (2020). Optimal estimation of Gaussian mixtures via denoised method of moments. *Annals of Statistics* 48, 1981--2007.

Zaffaroni, P. (2004). Contemporaneous aggregation of linear dynamic models in large economies. *Journal of Econometrics* 120, 75--102.